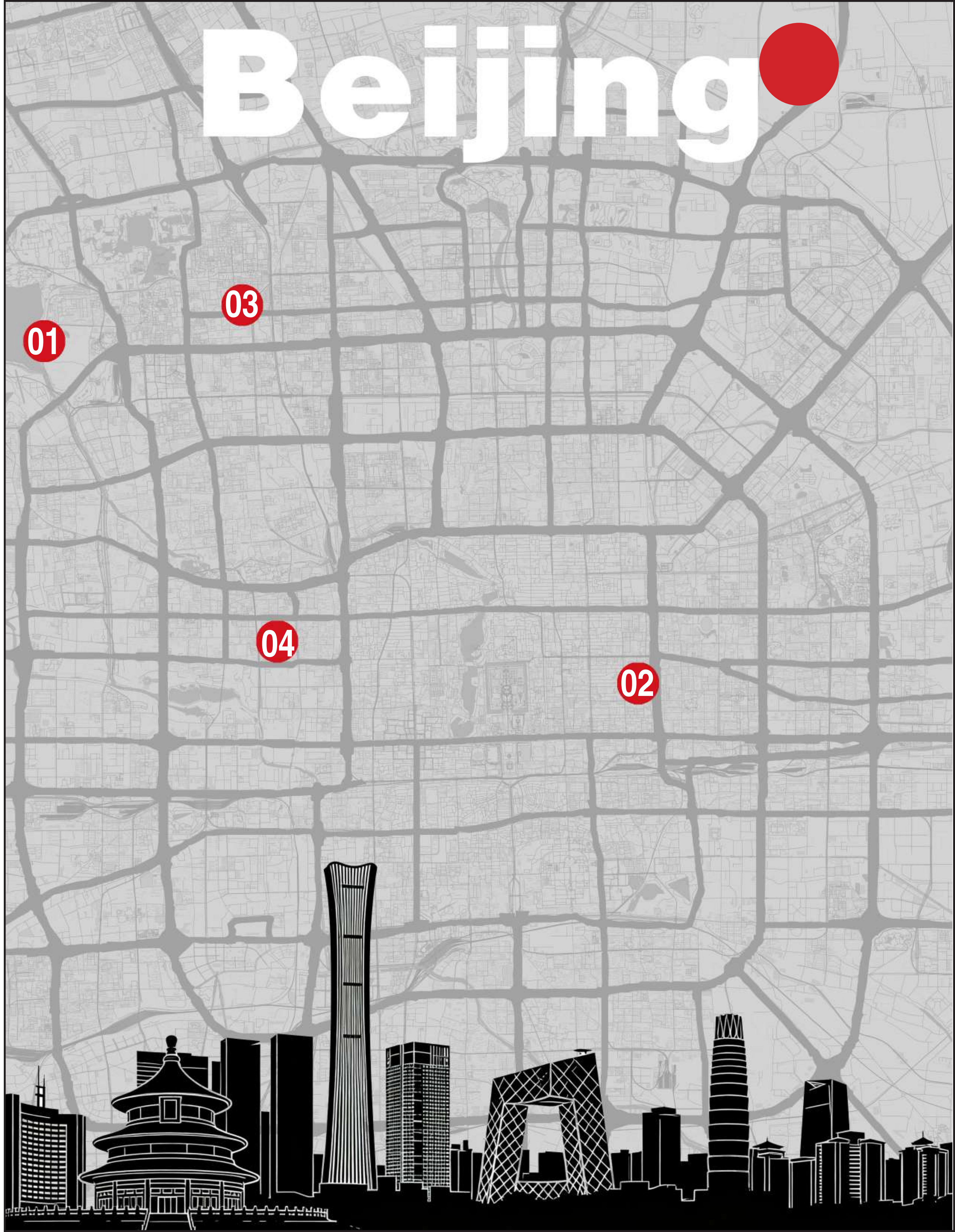
Transcend

Selected works 2024-2025

Bachelor degree in Architecture
Beijing University of Civil Engineering and Architecture

Wang Anji


Beijing
01
02
03
04

01 VERTICAL GARDEN

Instructor: Mr.Wang Tao
Spring, 2024
Individual Work
Project Status: Academic Project
Project Site: Summer Palace, Beijing, China.

As modern cities evolve toward vertical expansion, the horizontal layout of traditional gardens can hardly meet the needs of high-density human settlements. This design extracts the land textures and spatial topology of renowned Suzhou gardens, refines them into abstract geometric forms, and then reorganizes these forms with contemporary architectural vocabulary to construct a vertical garden standing beside the Summer Palace. This approach is not a mere replication of classical gardens; instead, at the intersection of historical contexts, it explores the symbiotic integration of traditional gardening wisdom and modern vertical living, allowing the millennia-old spirit of garden design to take on new forms in contemporary urban spaces.

Imperial Construction Inspired by Jiangnan Landscapes

276 years ago, Emperor Qianlong of the Qing Dynasty made multiple southern inspection tours, where the delicate and elegant landscapes of the Jiangnan region left a profound impression on him. This longing eventually translated into a construction project on the western outskirts of Beijing—the Summer Palace came into being. This imperial garden ingeniously drew on the exquisite layout of the "Su Causeway" and "Bai Causeway" of West Lake in Hangzhou, built the West Causeway on Kunming Lake, and accurately reproduced the graceful artistic conception of Jiangnan water towns. At the same time, it integrated the majestic momentum of northern imperial gardens, ultimately becoming a masterpiece that combines both "grandeur" and "elegance", serving as an exclusive retreat for emperors and empresses to escape the summer heat, handle state affairs, and enjoy sightseeing.

Draft design of the Summer Palace

Master Plan of the Summer Palace

Classical Charm and Technological Innovation

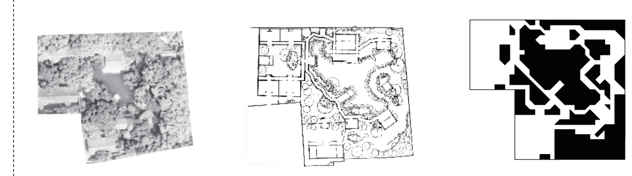
What is particularly fascinating is that Haidian District, where the Summer Palace is located, has risen to become a core engine of China's technological innovation in the past 20 years. The cluster of science and technology enterprises in Zhongguancun, the scientific research strength of top universities, and the R&D positions for cutting-edge technologies form a profound dialogue across time and space with the classical aesthetics of the Summer Palace here.

Precisely because of this, I have conceived the idea of reproducing the space of classical Jiangnan gardens—deeply integrating this classical charm with modern places such as work and residence. When workers devote themselves to R&D and innovation here, they only need to step out for a moment to enter the landscape scroll of the Summer Palace, find a resting place for inspiration in the infiltration of classical aesthetics, and let their tired thoughts slowly unfold amidst the lakes, mountains, pavilions, and terraces.

Stones in Chinese Classical Garden

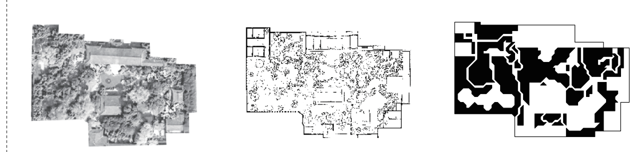
Typeplogy Analysis

In this research, nine classic Jiangnan gardens were chosen as the primary case studies, covering a diverse range of scales, styles, and historical contexts to ensure the comprehensiveness of the analysis. A rigorous abstraction methodology was adopted: first, the complex spatial configurations of these gardens were simplified by eliminating non-essential elements such as specific plant species, ornamentation, and material textures. Then, by mapping the built structures, open spaces, water bodies, and pathways, the underlying spatial figure-ground relationships were systematically identified, extracted, and documented, which provided critical insights into the traditional design philosophy that balances enclosure and openness in Jiangnan garden architecture.



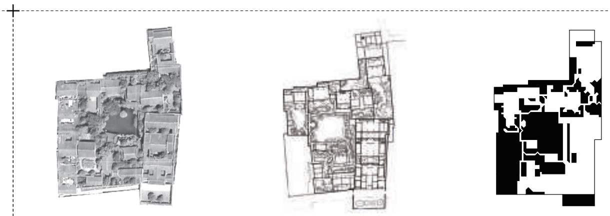
01 Tuisl Garden

It is a representative small Jiangnan garden of the late Qing Dynasty. Its name comes from Zuo Zhuan, reflecting the owner's introspection after retirement. Centered on water, it features a compact layout with rich spatial layers.



02 Geyuan Garden

A classic private garden of Yangzhou salt merchants in the Qing Dynasty, named after the owner's courtesy name and the abundant bamboo (the character "Ge" resembles bamboo leaves). Famous for its four-season rockeries made of different stones, creating distinct seasonal scenery.

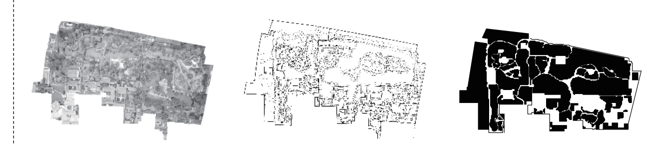


03 Master of Nets Garden

A delicate Suzhou classical garden, originally named "Fisherman's Seclusion". "Net Master" means fisherman, symbolizing the owner's longing for seclusion. Small in size but with a harmonious integration of water, land, pavilions and rockeries.

04 The Lingering Garden

One of Suzhou's four great classical gardens, its name implies "lingering in admiration". It is known for layered spaces and exquisite stone scenery, divided into four parts with diverse styles (water scenery in the center, buildings in the east, forests in the west, pastoral scenes in the north).



05 Humble Administrator's Garden

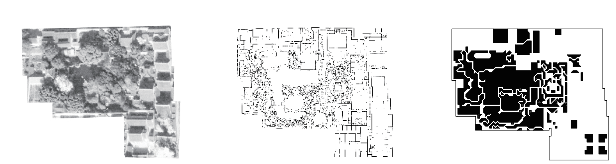
Suzhou's largest classical garden and one of China's four great classical gardens. Its name comes from Pan Yue's works, expressing the owner's self-deprecating seclusion. Dominated by water (1/3 of the area), it presents the free and natural charm of Jiangnan water towns.

06 Yongcui Mountain Villa

A mountain villa garden on Huqiu Mountain, Suzhou. "Yongcui" means "embracing emeralds", referring to its green mountain surroundings. Built along the terrain, it blends man-made structures with natural mountain scenery.

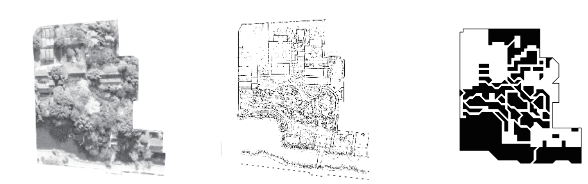
07 Jichang Garden

A historic Jiangnan garden at the foot of Huishan Mountain, Wuxi. Its name means "sending joy", reflecting the owner's enjoyment of natural scenery. It skillfully borrows the scenery of Huishan Mountain and Erquan Spring, with poetic mountain-water integration.



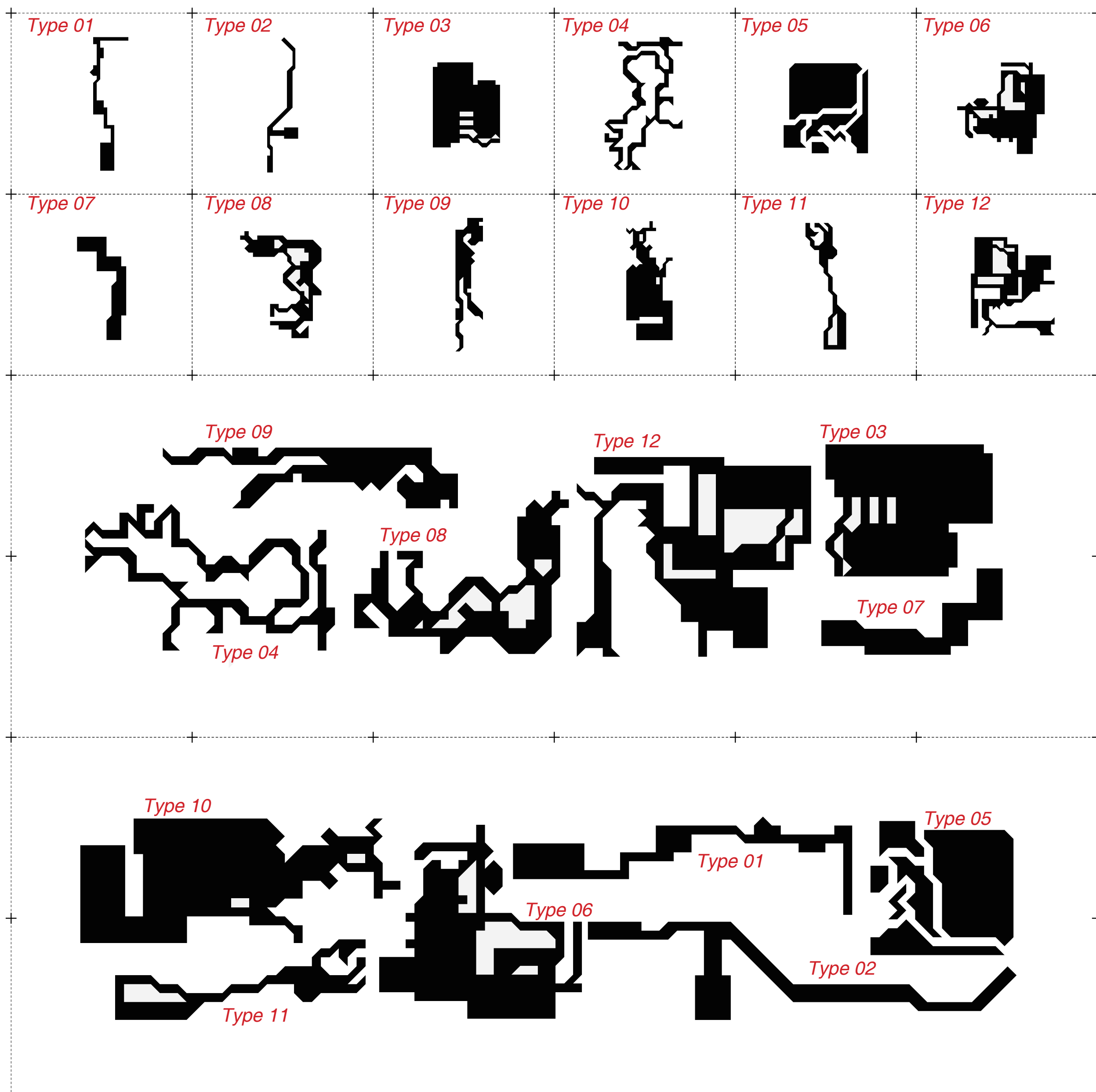
08 Lion Forest Garden

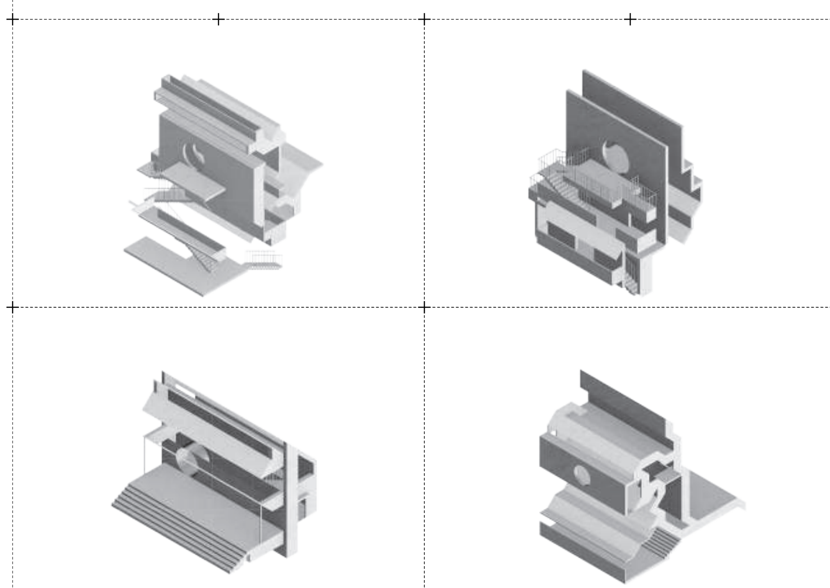
A Suzhou garden famous for labyrinthine Taihu stone rockeries shaped like lions (also linked to Buddhist symbolism). Its rockeries have winding paths, making it a unique "rockery garden" in Suzhou



09 Canglang Pavilion

Suzhou's oldest existing classical garden, named after Chu Ci, expressing the owner's indifference to fame. Built on a hill (unlike water-centered Suzhou gardens), it has a waterside corridor and borrows external scenery skillfully.





After reversing the figure-ground relationships, I extracted 12 distinctive abstract garden spaces, which were then subjected to permutation and combination within the constraints of the site area to explore diverse spatial layouts.

When conventional garden spaces are oriented vertically, unexpected vertical spatial effects are generated. These spaces are then further hollowed out and interconnected in the manner of Taihu stones, resulting in a spatial system that is exceptionally rich, engaging, and coherent. Functional zones are subsequently embedded dispersively within this intricate spatial framework.

Public Tea Space

"Chaotic System"

In classical gardens, the functions of space are both discrete and continuous. Discreteness ensures the pertinence of different usage needs, while continuity maintains the integrity of garden space and the smoothness of the visiting experience. I have also retained this feature in my design to create a composite space that is suitable for living, sightseeing and working.

Public Tea Space

Public Tea Space

Central Street

Public Space

02 Bicycle Community

Instructor: Mr. Shang Qian & MS. Jia Yuan
Spring, 2025
Individual Work
Project Status: Academic Project
Project Site: Chaoyang, Beijing, China

This project takes the dual meanings of "cycling" as its core conceptual anchor, deeply integrating the spatial and functional attributes of "bicycling" with the sustainable design philosophy of "recycling", so as to create a multifunctional community space threaded by bicycles and supported by reusable steel structures.

The combination of the two elevates "cycling" from a mere spatial design thread to a metaphor for a community lifestyle—bicycles weave together the daily routines of residents, while recyclable steel structures underpin the sustainable evolution of the space. Ultimately, it builds a low-carbon, vibrant and symbiotic new community model, exploring the balanced coexistence of function and ecology, humanity and technology.

Beijing's Bicycle Culture

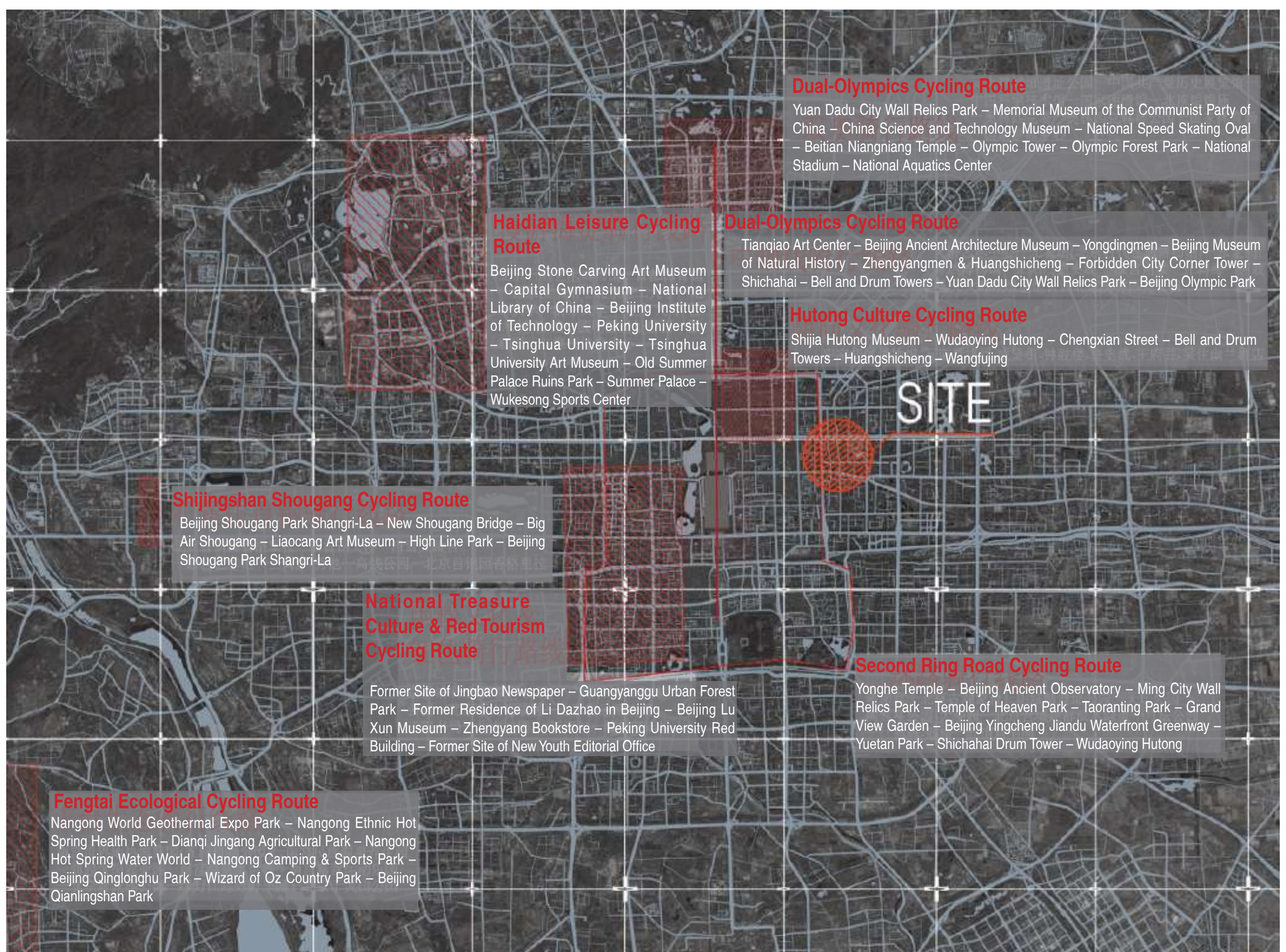
Known as the "Capital of the Bicycle Kingdom", Beijing boasts a unique bicycle culture. In the 1970s and 1980s, bicycles were among the "Four Big Items" (symbolizing family prosperity), and the morning and evening "bicycle tides" were iconic urban scenes. Today, with the promotion of low-carbon travel, dedicated bicycle lanes have been increasingly improved. The popularity of shared bikes has revitalized bicycle culture, making it a preferred choice for short-distance trips.

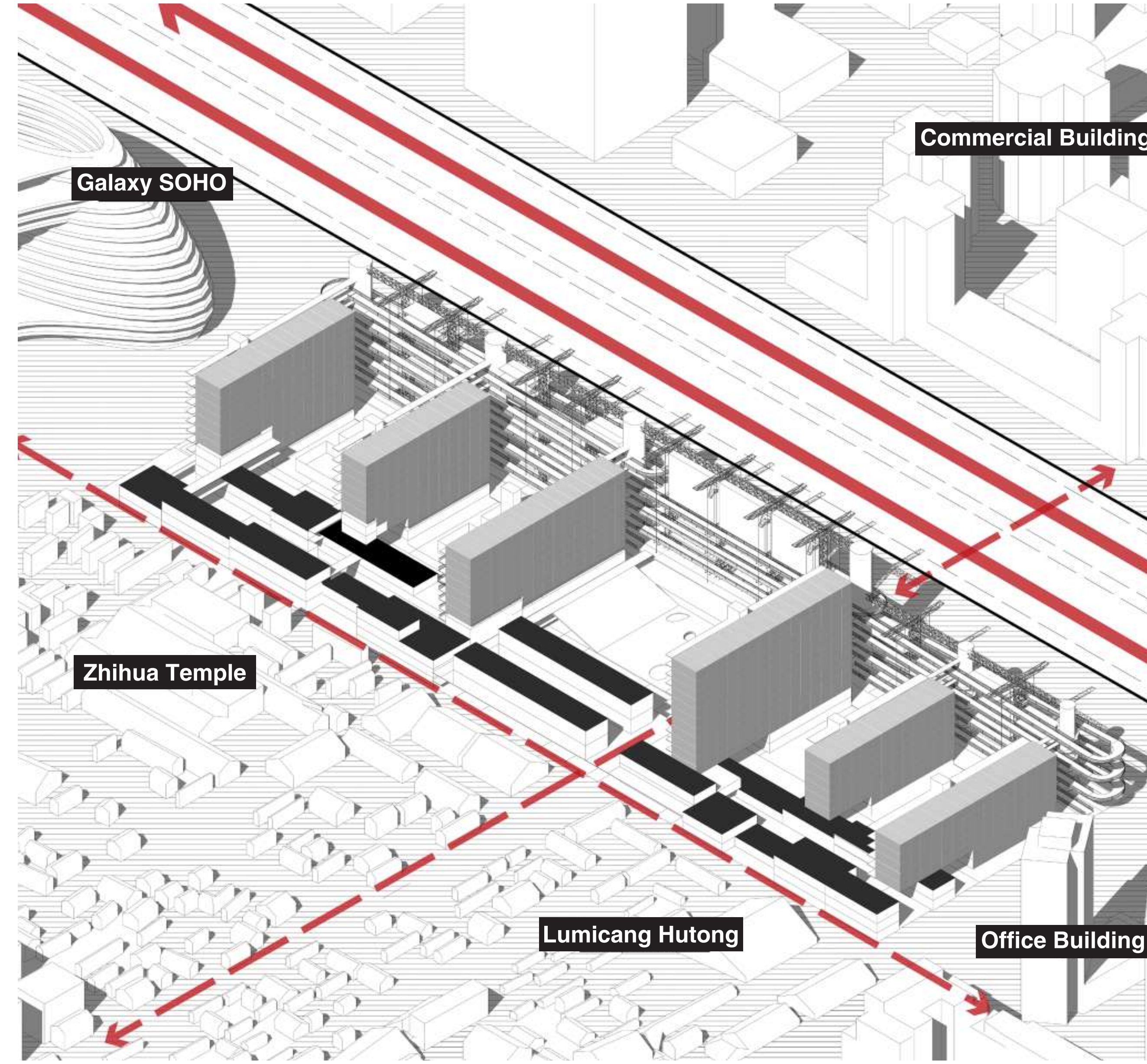
Cycling is also an excellent way for tourists to explore Beijing. Riding through hutongs allows them to experience local life, while cycling along Chang'an Avenue lets them appreciate the blend of history and modernity.

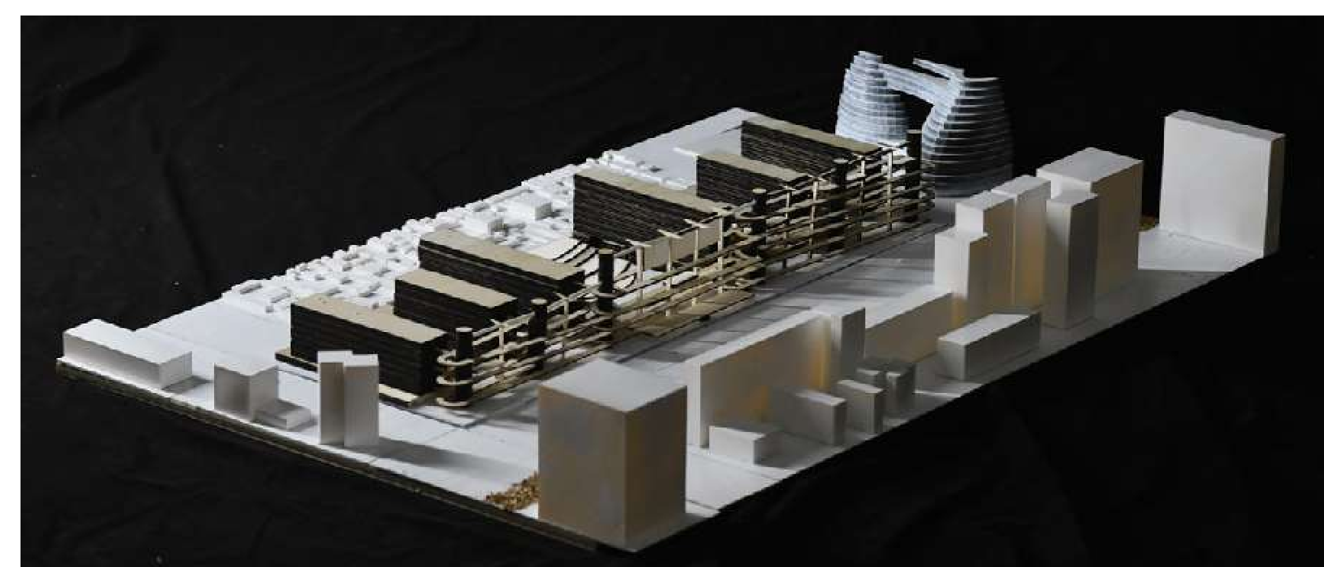
Various cycling support measures and activities reflect the city's pursuit of sustainable and livable development, and bicycles have always been an important part of Beijing's story.

Urban Function

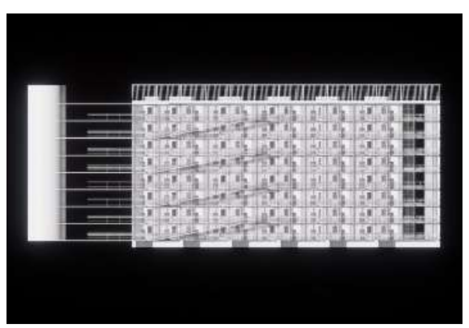
From an urban design perspective, the site is adjacent to a host of classic cycling routes in Beijing. Hence, I decided to develop the community into a city-wide cycling hub, where the cycling-friendly environment will be meticulously crafted with a full range of facilities and activities to attract cycling enthusiasts from across the city.





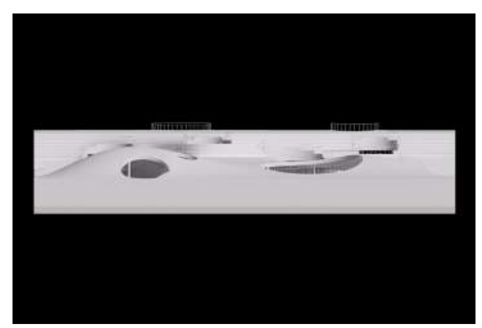


Residential Building



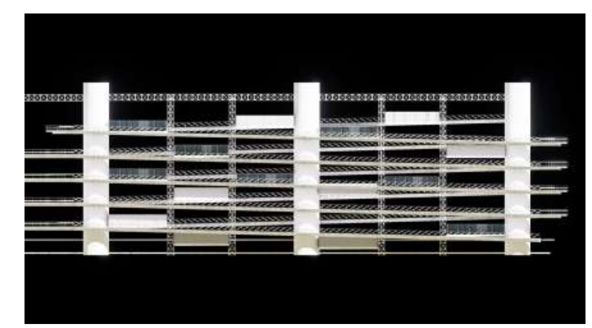
Activity Space

Earth-sheltered Space

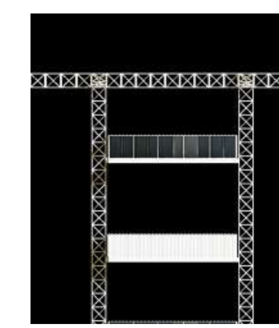


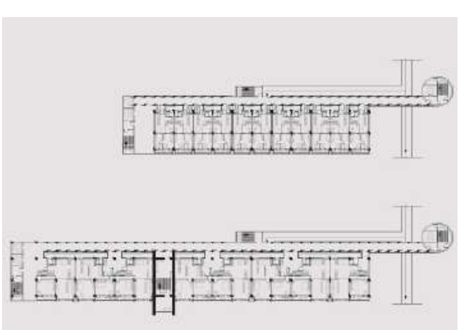
Aerial View

Suspension Bicycle Frame

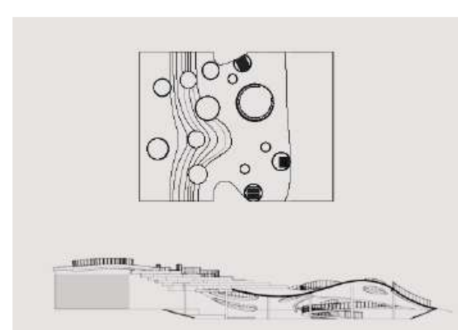


Elevator Stores

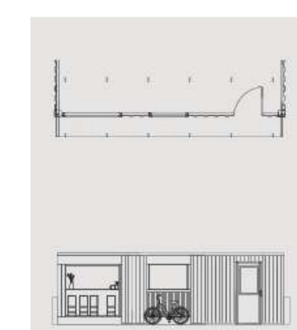




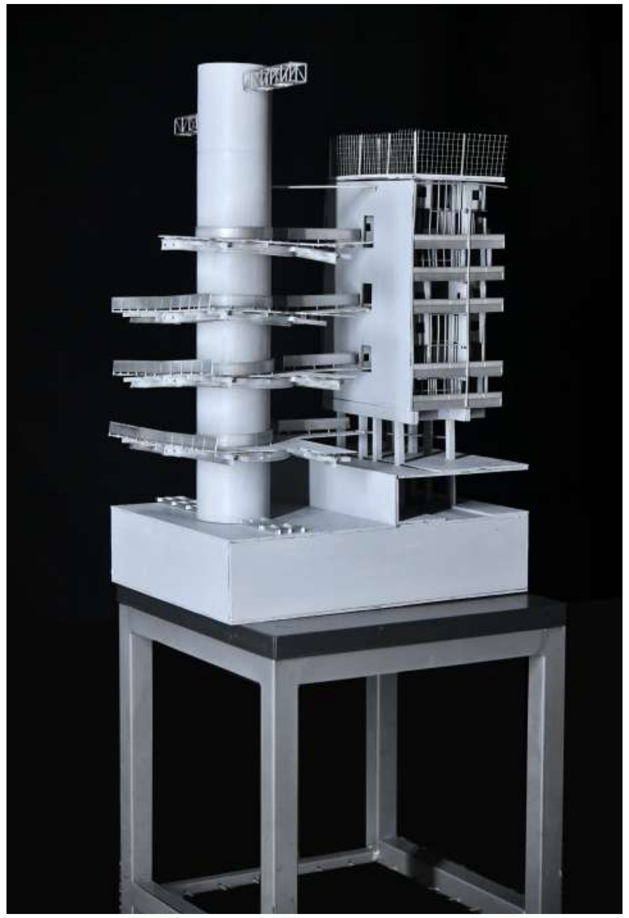
Residential Building



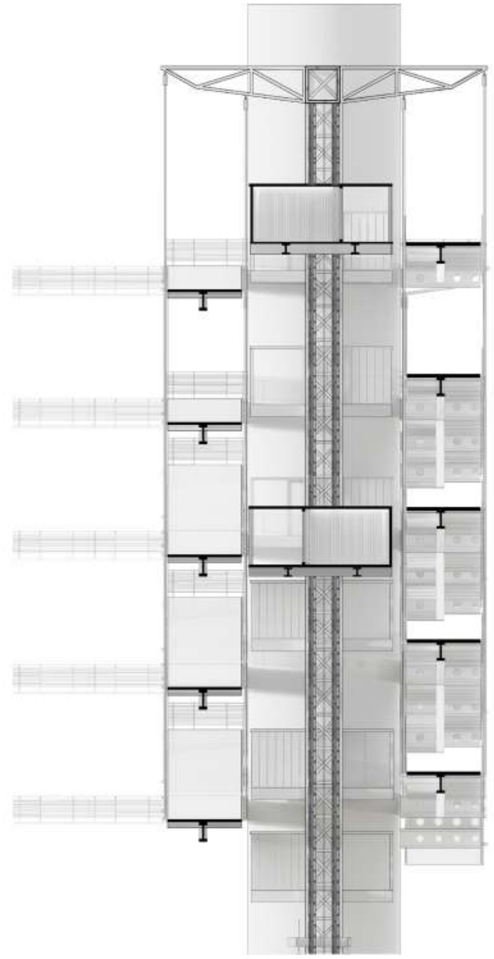
Subterranean Garden



Suspension Bicycle Frame



Model

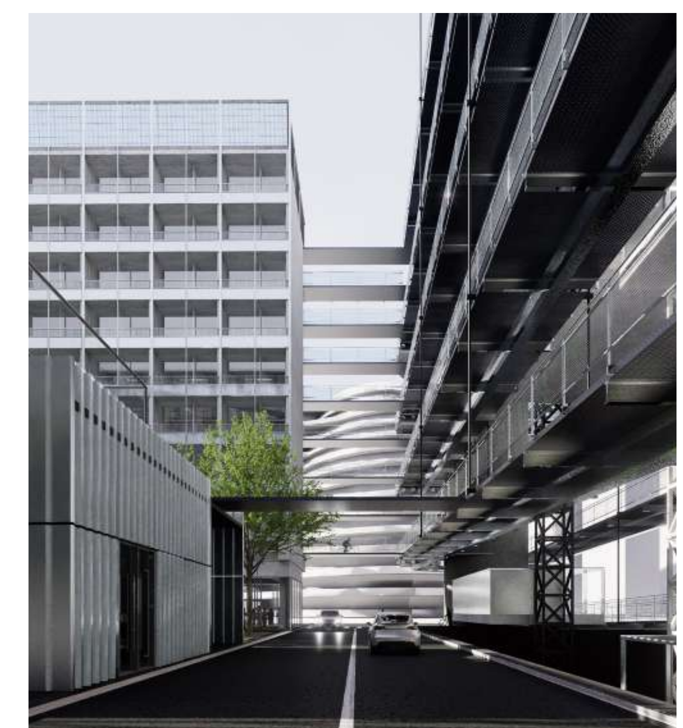


Spatial Distribution attern

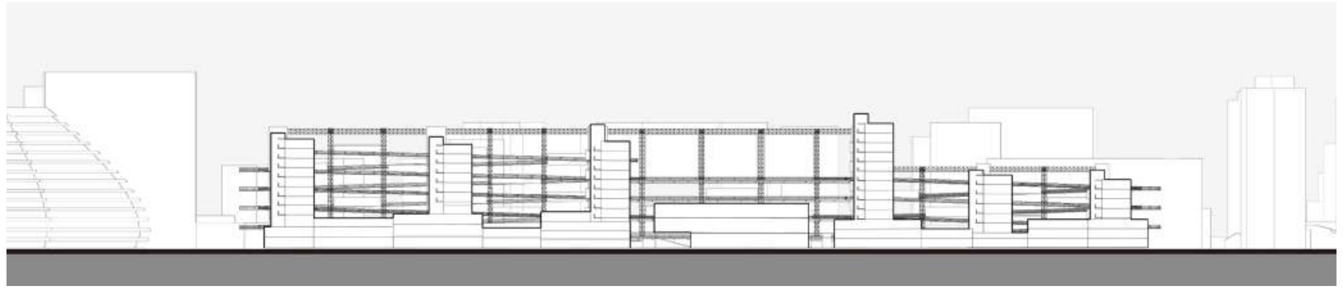
Spatial Distribution attern

Cycle Track

As a bicycle-friendly neighborhood, we have constructed this massive rack system to enable residents to cycle directly to their doorsteps and ride up to the upper floors. Elevator cages are installed in the middle of the racks, which can be repurposed as retail units. These cages can be upgraded to adapt to the varying height differences of the bicycle lanes, and are also detachable to allow real estate developers to add or remove suspended shops as needed. Cyclists can park their bikes right at the shop entrances during their ride while enjoying the scenery from a higher vantage point.

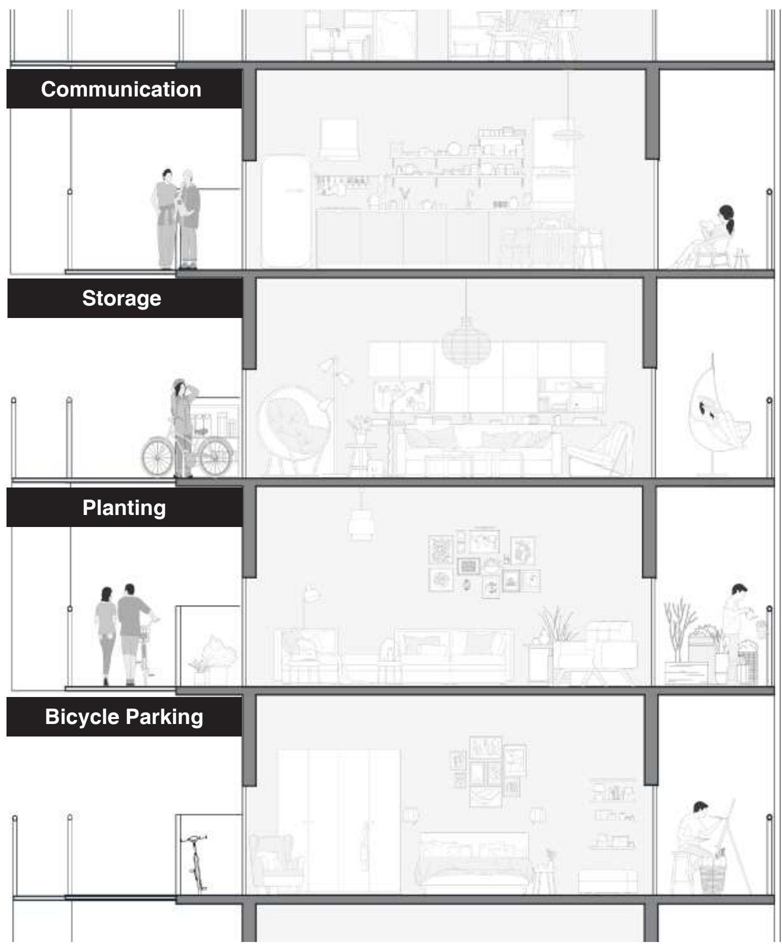


Motoeway



Long Side Section

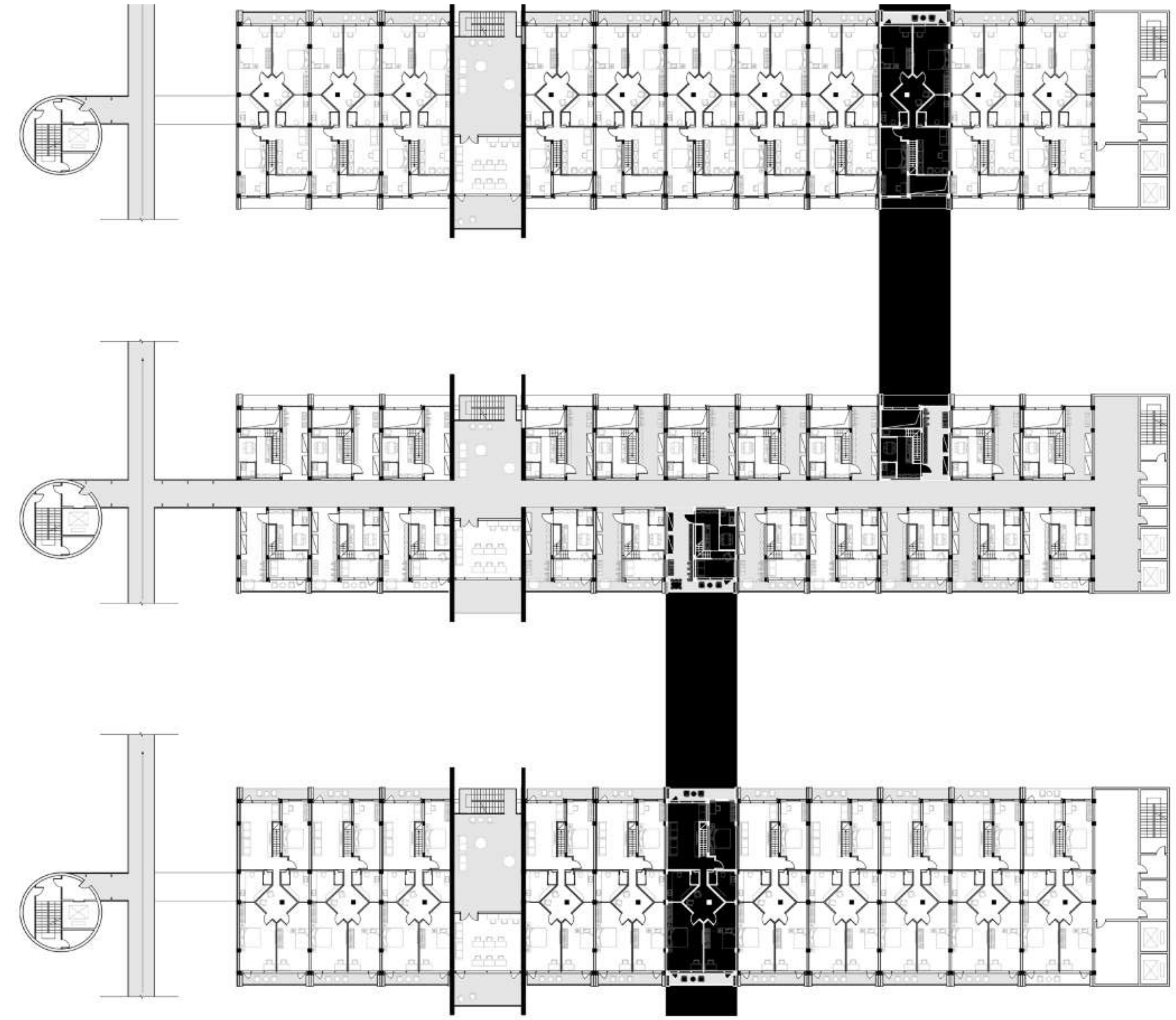
The six buildings aligned in a south-to-north sequence are seamlessly linked by an elevated bicycle path, which not only serves as a functional traffic artery for cyclists but also acts as a social corridor—greatly simplifying the daily visits between residents and fostering a closer neighborhood bond. In addition, a series of suspended shops are ingeniously integrated into the design along the path; this thoughtful layout enables residents to access their desired goods and daily necessities conveniently without the need to go downstairs, thus optimizing the efficiency of daily life within the community.



Residential Building Section

Residential Building Elevation

To ensure that bicycles can be ridden directly to the residents' doorsteps while also preserving the daylighting needs of the households, I arranged the bicycle lanes on the north side, leaving the south-facing side of the rooms unobstructed for natural light. A sufficient area is allocated between the bicycle lanes and the residential entrances, which serves multiple functions such as bike parking, storage, and planting.

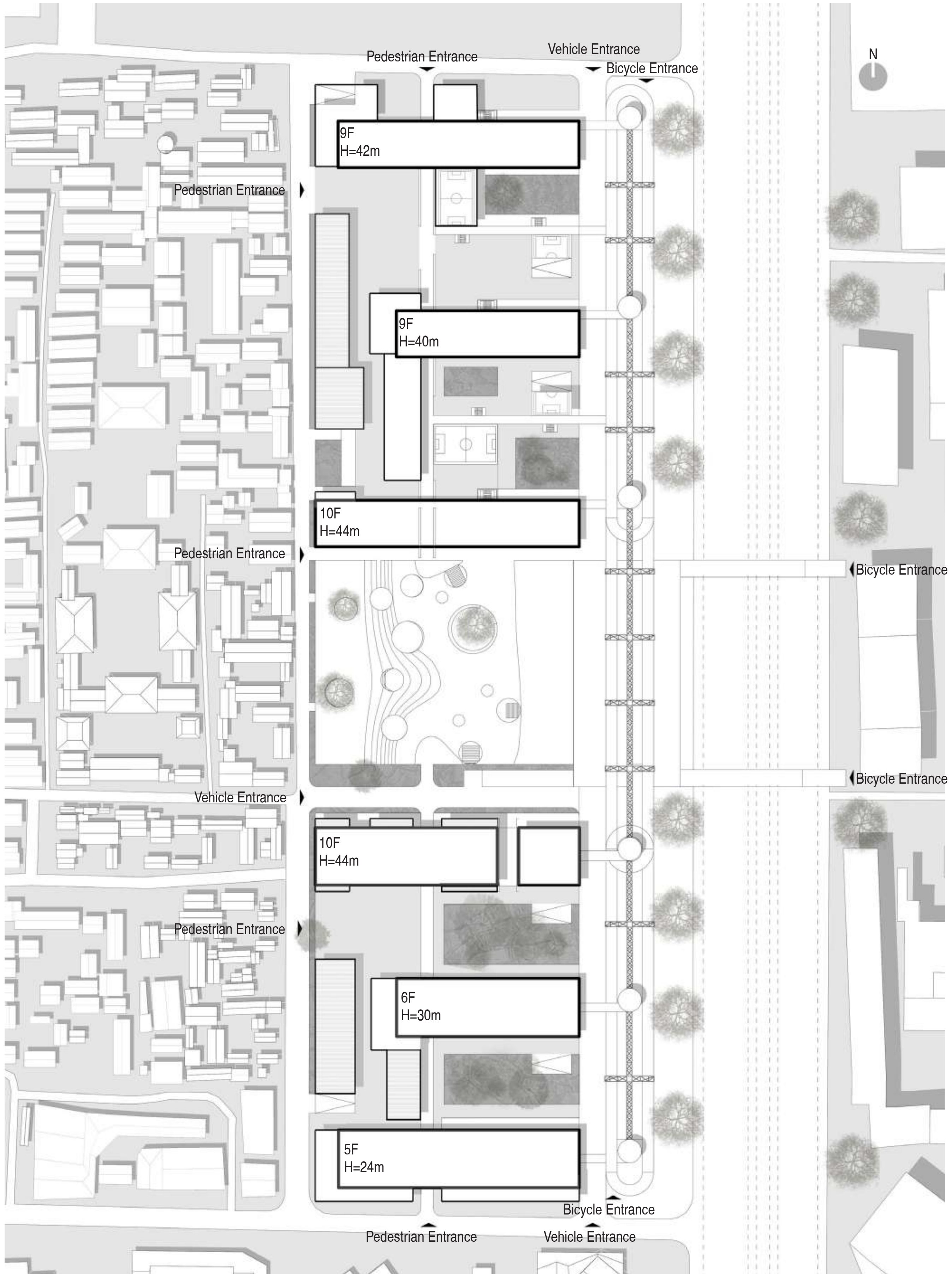


Residential Foor Plan

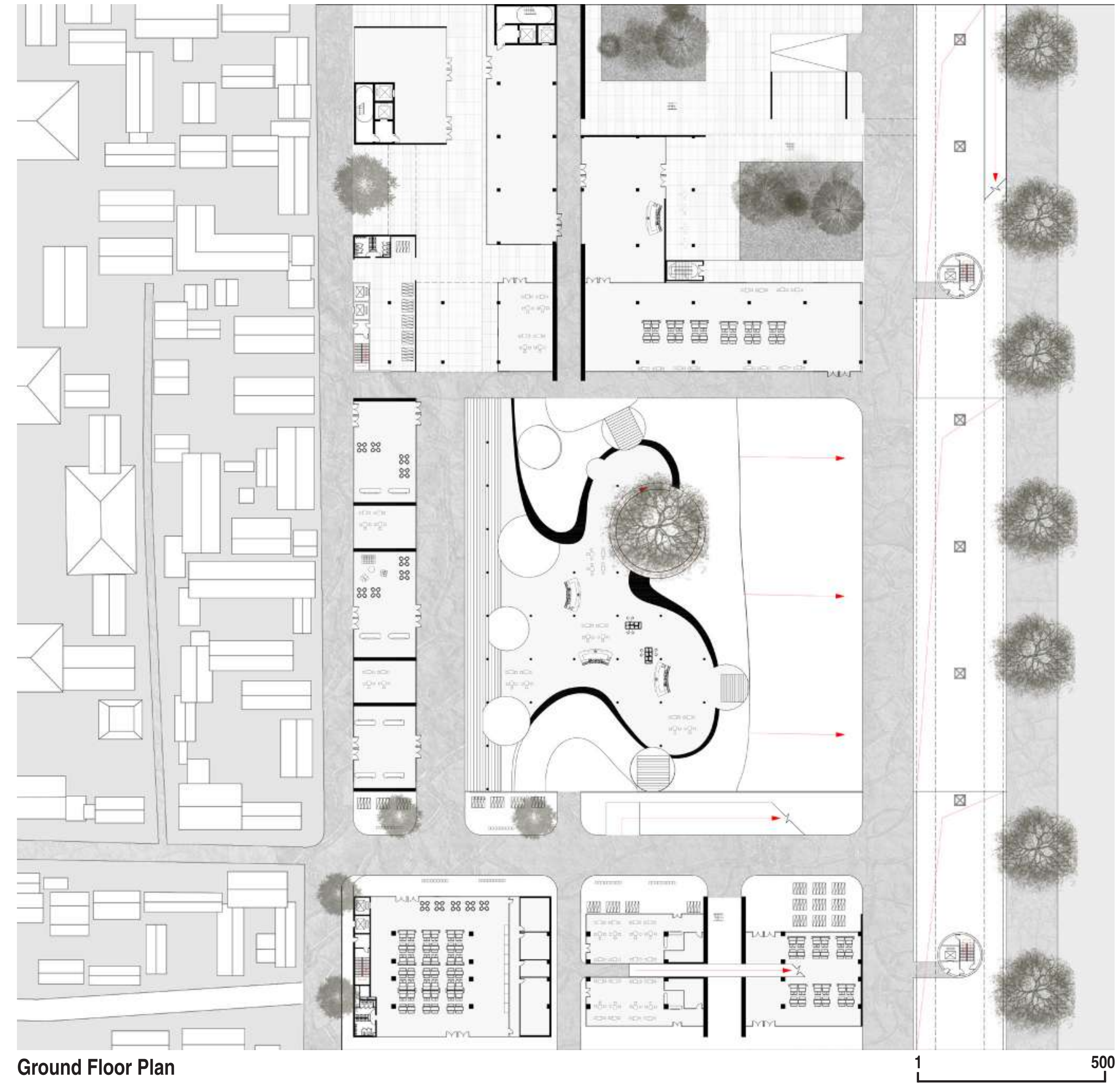
To further diversify residential options, I have also designed loft-style units within the complex. A distinctive feature of these units is that the elevated bicycle path runs directly through the middle of the buildings. Despite this integrated traffic design, the loft layout is meticulously optimized to ensure that every household enjoys unobstructed south-facing daylighting, perfectly balancing functional connectivity and living comfort.

Residential Building Corridor

Activity Space



Masterplan



Ground Floor Plan

Typical Block

Earth-sheltered Space

In the central area of the entire residential zone, we have designed a large earth-sheltered activity building adjacent to the public road. On the ground level, people can sit casually against the frames to watch movies, and bicycles can ride along the sloped surfaces. Trees emerge from the ground through openings, which also provide natural lighting for the underground public spaces.

03 AUTONOMOUS VEHICLE INCUBATOR

Instructor: Mr. Xu Yuejia
Winter, 2024
Individual Work
Project Status: Academic Project
Project Site: Zhongguancun, Beijing, China

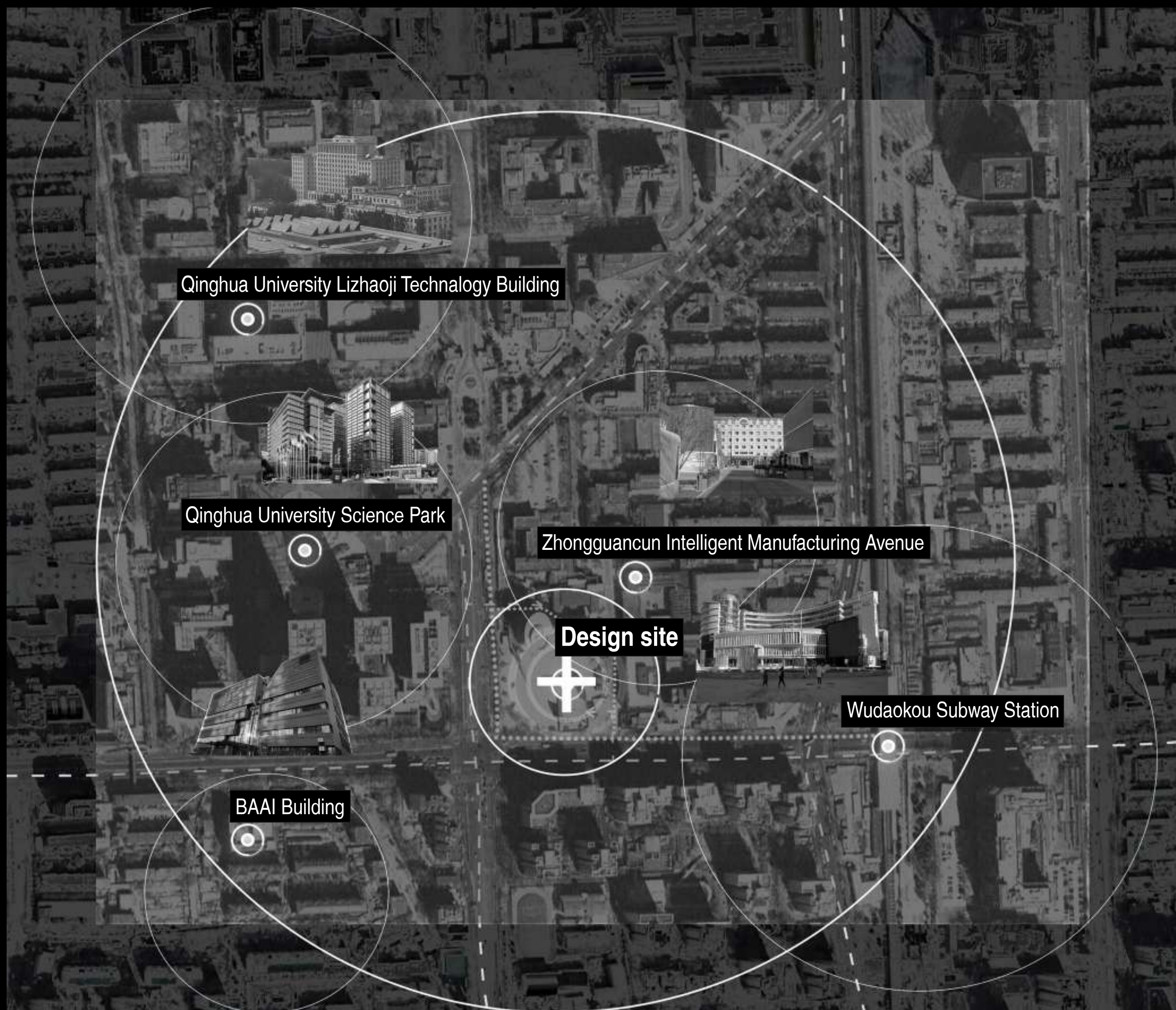
This project is launched to explore how emerging AI image generation technologies can be applied in design. The architectural goal here is to transform Dongsheng Building—situated in Beijing's tech core district—into an incubator tailored to a specific new industry. After researching a range of cutting-edge high-tech sectors at the forefront of today's tech landscape, I settled on the autonomous driving vehicle industry.

As technologies like artificial intelligence, large language models, and telecommunications continue to advance rapidly, autonomous driving technology will see widespread adoption across China in the coming years. It will be deployed in areas such as public transit, logistics, and ride-sharing, spanning use cases including Robobus, Robotaxi, port operations, manufacturing park logistics, mining site transport, sanitation services, long-haul freight, and last-mile delivery.

These application scenarios not only showcase the vast potential of autonomous driving technology but also provide concrete guidance for my architectural design—from functional requirements to spatial layout.

Analysis of Surround Sites

After deciding on the design of an incubator for the autonomous driving industry—a burgeoning sector brimming with boundless potential and innovative dynamism—as my core theme, I proceeded to analyze whether the target location boasted geographical advantages for constructing such an incubator. My focus was directed at the areas surrounding the site, where I conducted meticulous research and collation on universities, research institutions, and R&D centers of relevant enterprises that could serve as vital talent pools for the autonomous driving field. I also explored in detail their talent cultivation systems, scientific research capabilities, as well as the scale and characteristics of talent output.



Site Issues

Problem1

Traffic congestion is acute at the southwest corner of the site. During peak hours, crowds converge here in a flood, with pedestrians jostling shoulder to shoulder and shuffling slowly in the limited space. A mix of vehicles compounds the chaos: bicycle bells ring constantly, e-bikes weave cautiously through the throng, and motor vehicles honk repeatedly to force a way through. Without proper traffic diversion facilities, conflicting flows of pedestrians and vehicles frequently cause gridlock, plunging the already busy area into disorder. Prolonged congestion not only delays commuters but also heightens risks of collisions and falls, creating major inconvenience and potential hazards.

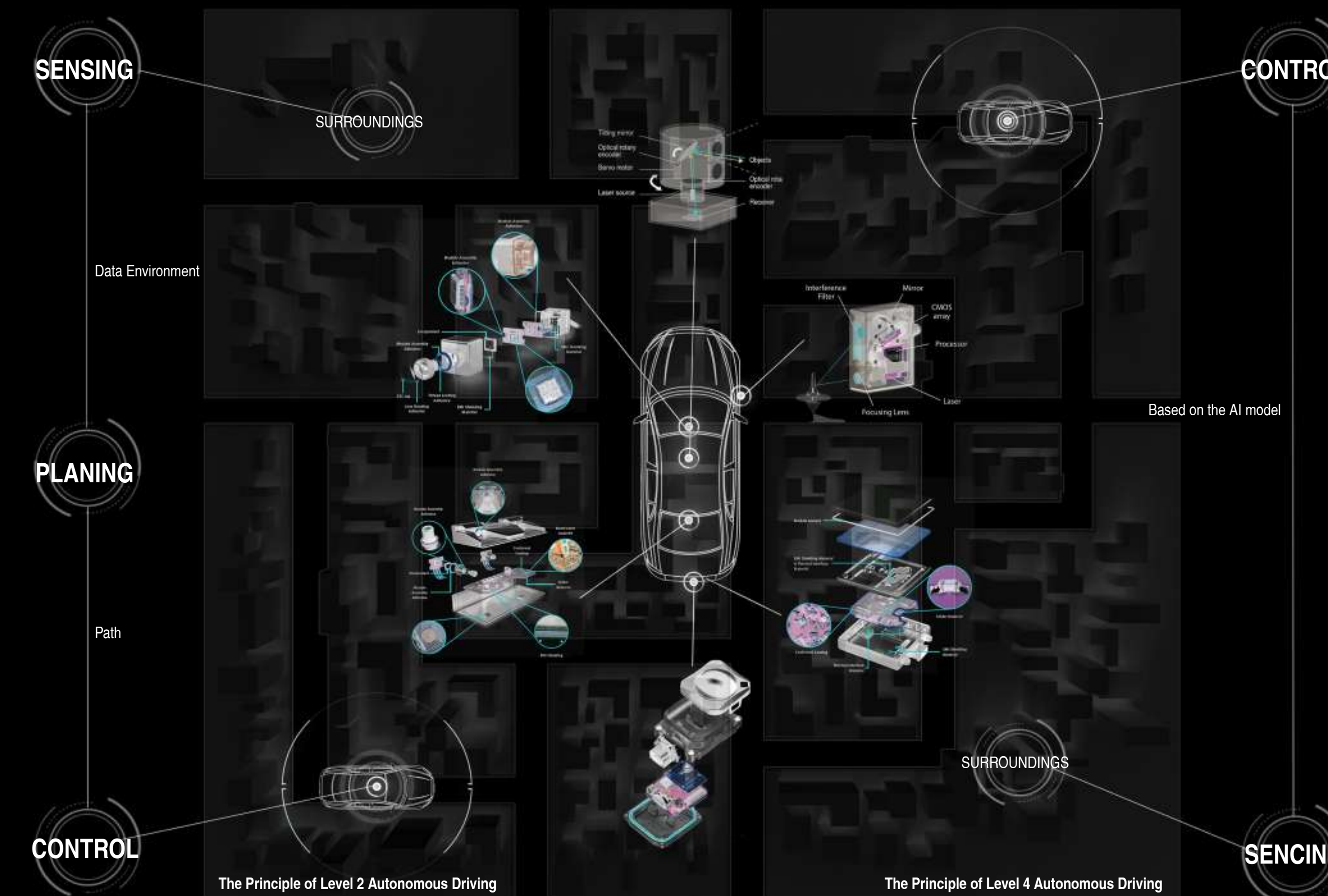
Problem2

On the north side of the site, there stands a non-removable boiler house which is in close proximity to the main building. Coupled with the parking spaces designated on the north side, this arrangement has led to relatively congested traffic in the area. What is more, the overlapping of pedestrian flow and vehicle traffic flow here poses certain potential safety hazards. Currently, there are some food and beverage establishments on the north side of the building, but they are obviously lacking in vitality with a relatively small number of diners. In addition, the tall buildings on both sides result in poor lighting conditions and create wind tunnels, which will exacerbate the impact of strong winds on pedestrians when they pass through on blustery days.

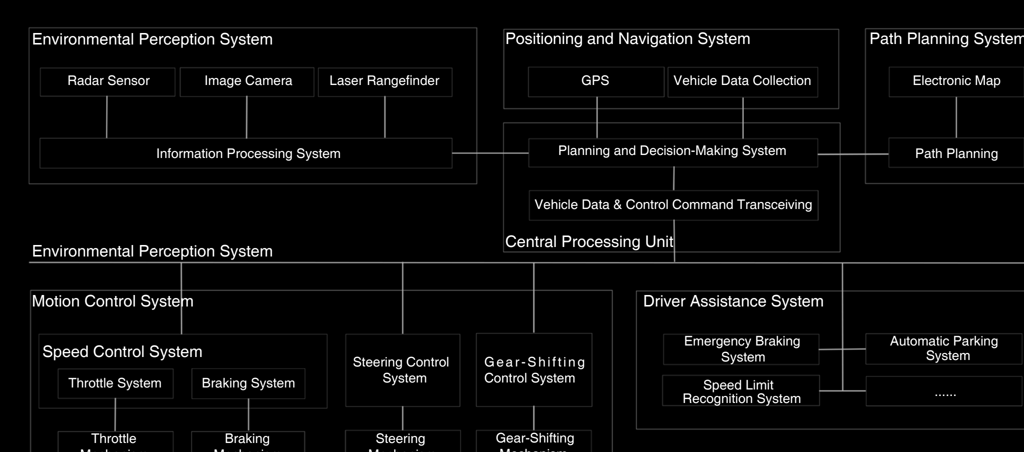
Problem3

The street vitality between Dongsheng Building and Zhongguancun Intelligent Manufacturing Street is rather lackluster. Moreover, most of the road space is occupied by food delivery e-bikes and other non-motor vehicles, which disrupts the normal passage of pedestrians. In addition, the eastern side of Dongsheng Building has a very limited functional value; currently, only a single café operates there, which attracts a small number of visitors and thus plays a negligible role in enlivening the area.

Autonomous Driving Theory



Upstream Technical Points of the Autonomous Driving Industry



Analysis Objective

To design rational and functional buildings tailored to the autonomous driving vehicle industry, I conducted in-depth research on the industry's background information and summarized the corresponding spatial requirements.

Conclusion

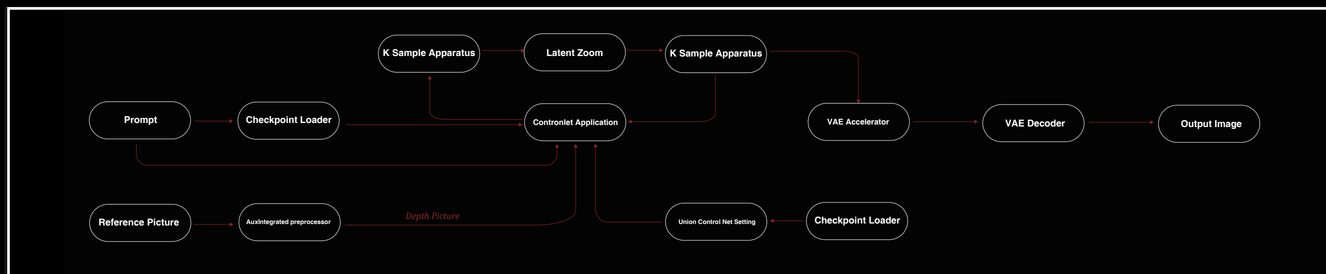
To support the innovation and industrialization of autonomous driving technologies, the incubator must integrate R&D laboratories, simulation & testing zones, collaborative workspaces, and talent service facilities, with a flexible spatial layout that adapts to technological evolution and industrial chain collaboration needs.

Artificial Intelligence-Assisted Design

During the first phase of the project, which involved AI image generation software-assisted design, my goal was to integrate elements of autonomous driving—a cutting-edge technology—into my architectural design. I first conducted a series of brainstorming sessions to explore the characteristics of autonomous driving technology, such as smooth dynamics, seamless connectivity, and intelligent interaction, all of which I aimed to embody in the design.

To capture these concepts, I decided to adopt an extensive use of curved lines in the draft sketching process. Curves not only mimic the smooth movement trajectories of autonomous vehicles but also convey a sense of futurism and dynamic beauty. Leveraging the intelligent drawing tools in the AI image generation software, I effortlessly created complex yet elegant curves. These curves not only form the outline of the building but also become an integral part of the interior design, fostering a fluid spatial experience.

"Stable Diffusion ControlNet Assisted Generation" Workflow



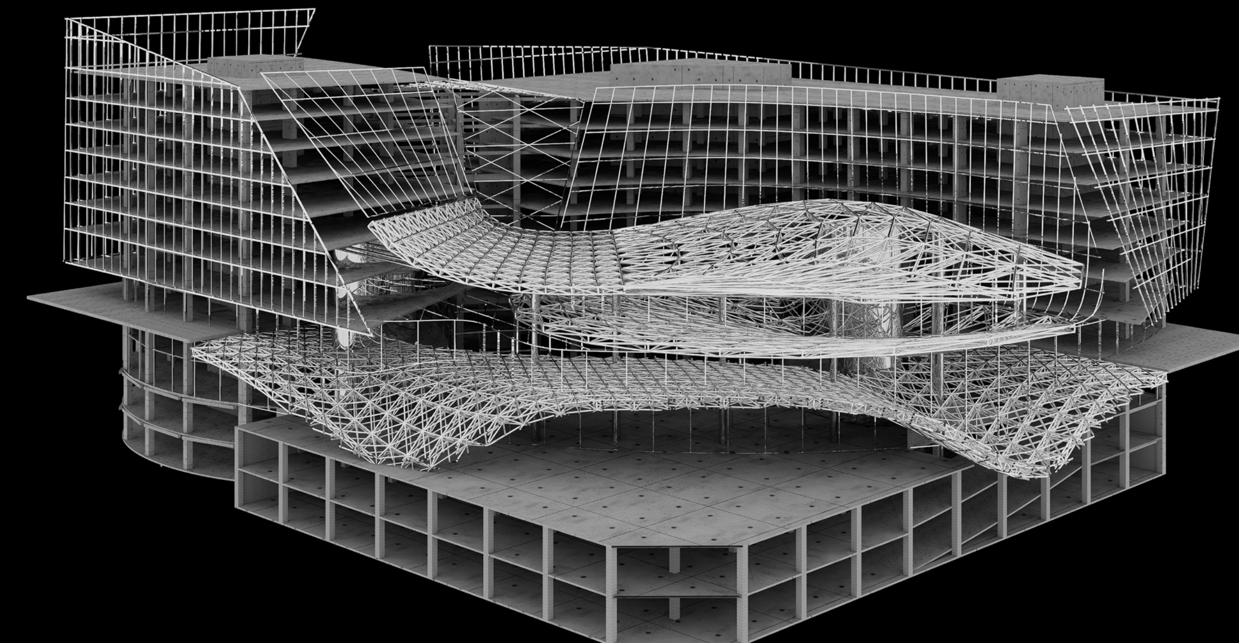
Original Picture

The Choosing Image

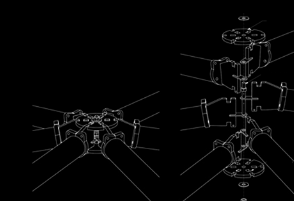
Out Put Images

Completed Design

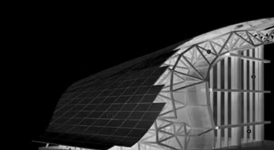
Construction and Details



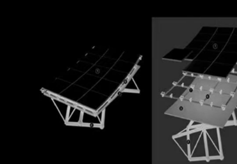
Construction



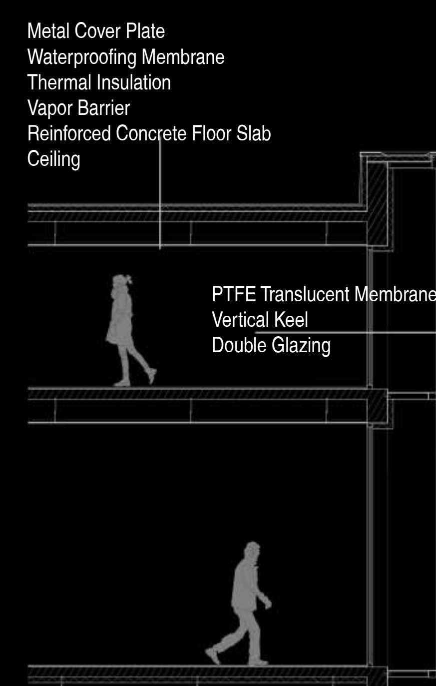
Connection Node



Section of Grid Structure

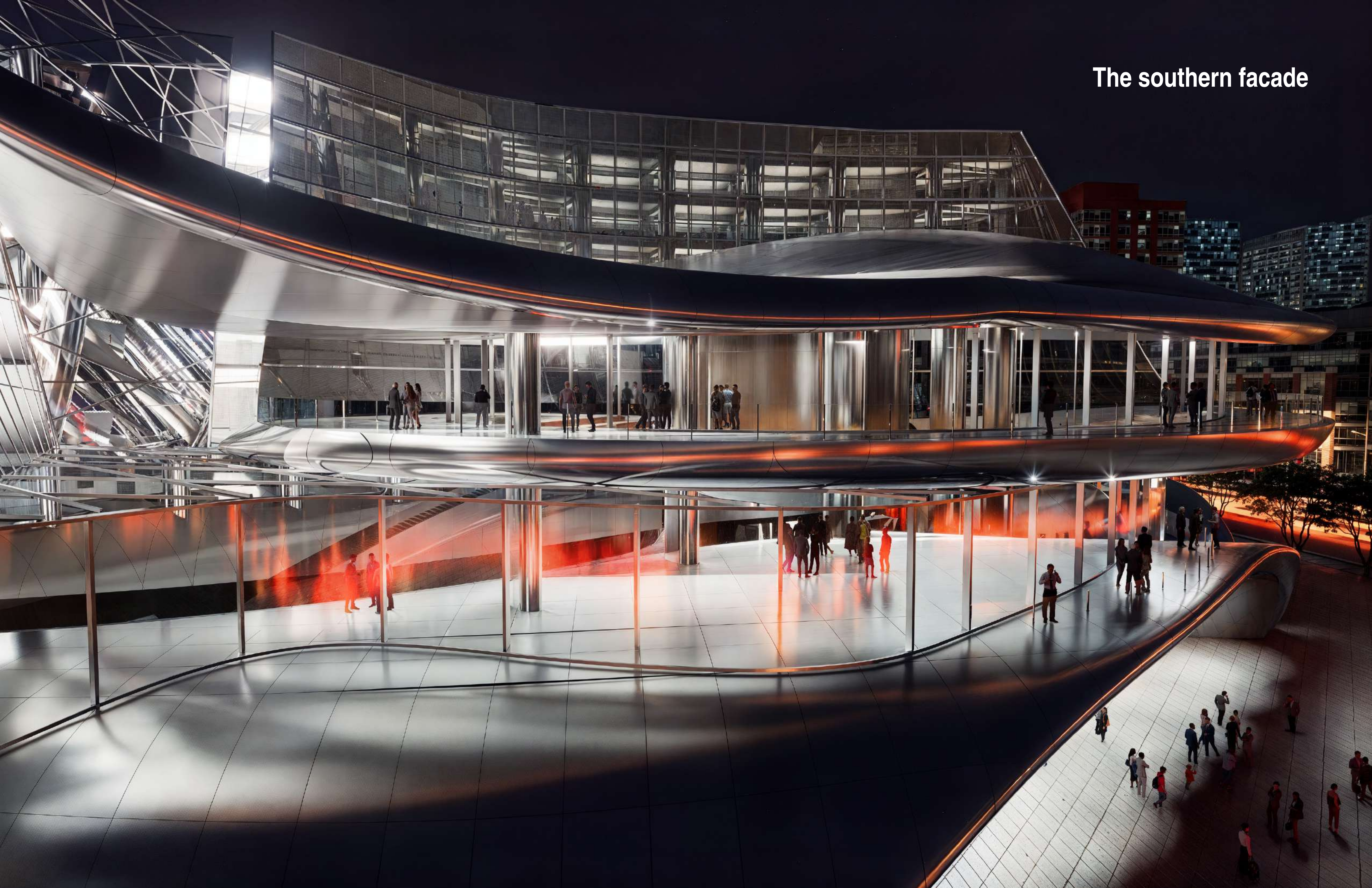


Grid Structure Skin



North Facade

Completed Design

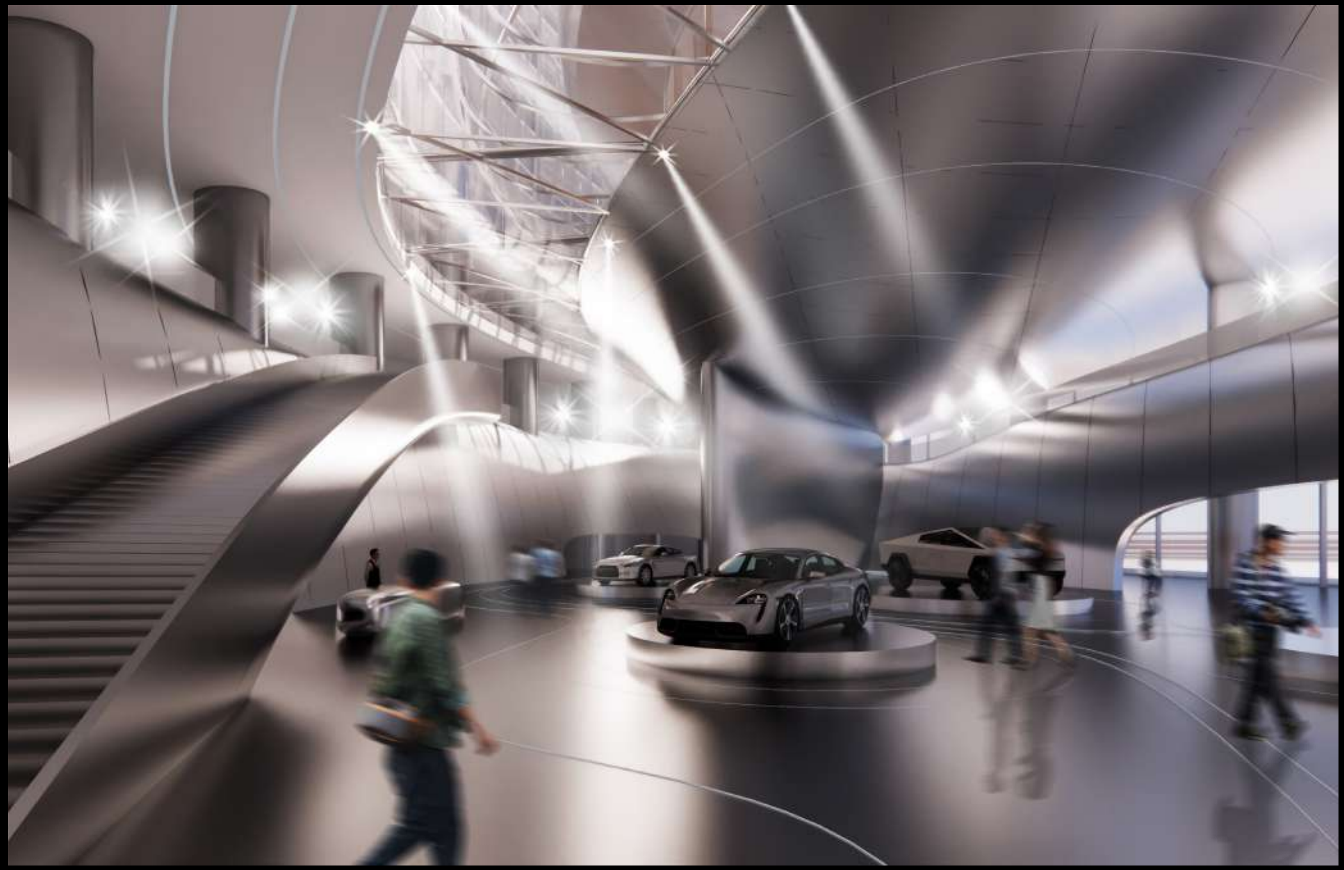


The southern facade

Office Space

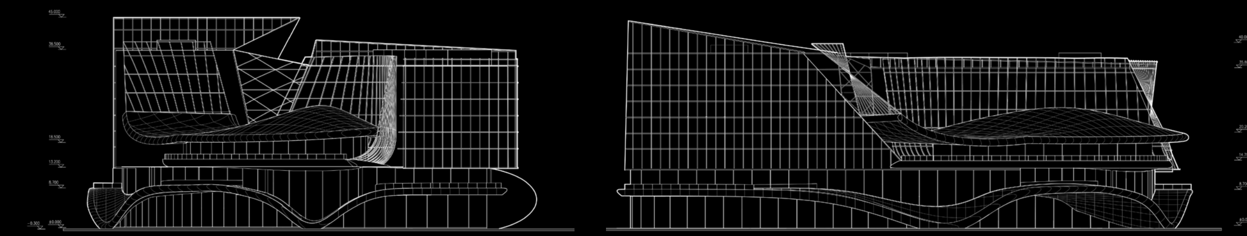
Exhibition Hall

Press Conference Venue

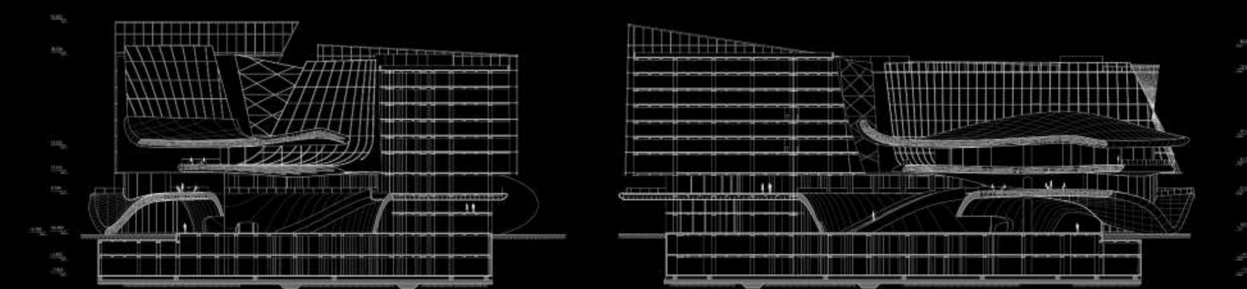


Exhibition Hall

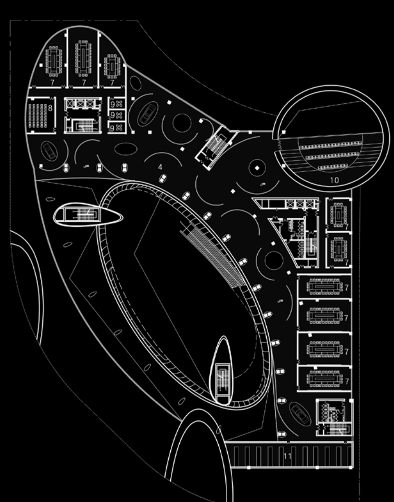
Spuare



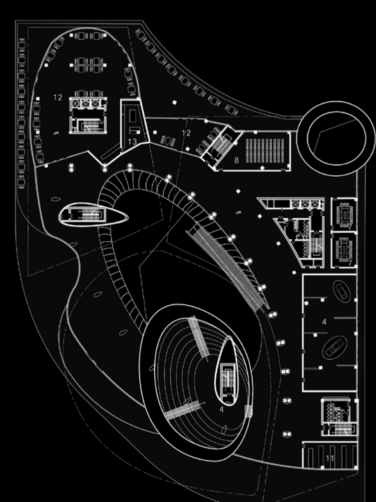
Elevation



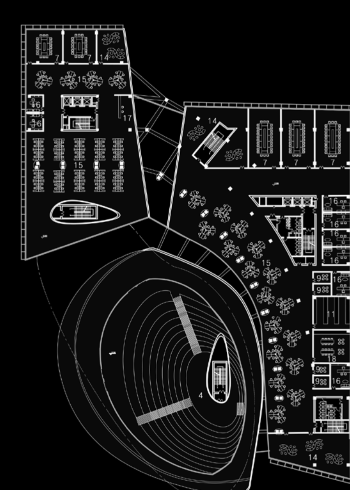
Section



Ground Floor Plan



Second Floor Plan



Third Floor Plan

Square

FUSUIJING HAUNTED HOUSE

04 FUSUIJING HAUNTED HOUSE

Instructor: Ms. Sun Yanchen
Winter, 2023
Individual Work
Project Status: Academic Project
Project Site: Xicheng, Beijing, China

As one of the four most famed "haunted buildings" in Beijing, Fusui Jing Building has stood majestically in the very heart of the low-rise, maze-like hutong district for decades. Though now long abandoned and left to the quiet ravages of time, the myriad stories and legends shrouding it still linger as a favorite topic of lively conversation among locals. Centered on the innovative approach of spatial translation of films from four relevant historical periods, this design seamlessly integrates a diverse array of new functions—including exhibition halls, dynamic livehouses, immersive theaters, and thematic museums—into the building's original structural framework. It seeks not only to shatter the barriers of historical information isolation, guiding people to step back into that once unspoken and forbidden-to-discuss special era through multidimensional perspectives of architecture, narrative, and emotion, but also to inject a vibrant new lease on life into this weathered, time-honored structure.

Beginning

Once Beijing's first residential building fitted with elevators—a coveted address that countless people dreamed of calling home—Fusui Jing Building now stands dilapidated and vacant. It has even earned a spot on the list of the "Four Haunted Buildings in the Capital" thanks to its eerie folk legends, and served as the inspiration for multiple film adaptations. What untold story lies behind this structure? Let us now lift the veil on its mysterious past.

Haunted Room

Related Movie

Picture from 1958

1958 | 1967 | 2005 | ......

Great Leap Forward

Constrution

Constructed in 1958 as a product of the Great Leap Forward, the Fusui Jing Building—once a renowned landmark also called the "People's Commune Building" and "Communist Building"—ceased operation in 2005. A hallmark of that era was its communal canteens, where the collective "common pot" dining practice was widely promoted. Residents admitted to the building had to pass strict political screening; with about 40 households per floor, it housed 358 families in total.

Cultural Revolution

Be Haunted

It was rumored that in 1967, a lady living in the building committed suicide by jumping off the building under immense pressure due to her "alleged moral misconduct". Since then, tales of hauntings have circulated. The controversial memories left by that specific historical period lacked official channels for public discussion and clarification. As a result, ordinary people vented their complicated emotions toward the past indirectly by making up tragic stories associated with the building, which thus became a vessel for emotional projection.

Present

Be Abandoned

As time went on, these unsubstantiated tales drifted further away from historical facts, and "haunted building" became a synonym for the structure. Especially after the building was vacated and left unused in 2005 due to fire risks, its dilapidated state from years of neglect lent greater credibility to the rumors. Eventually, the moniker "Ghost Eight-Story Building" replaced its original name "Fusui Jing Building" and became its most widely recognized identifier.

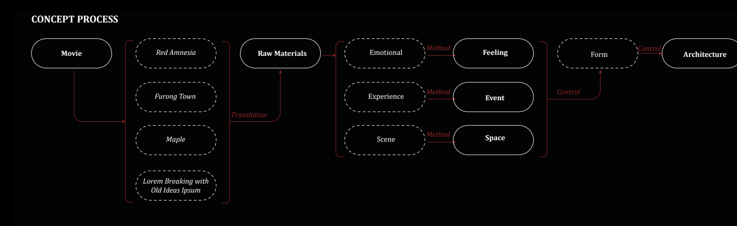
After the renovation

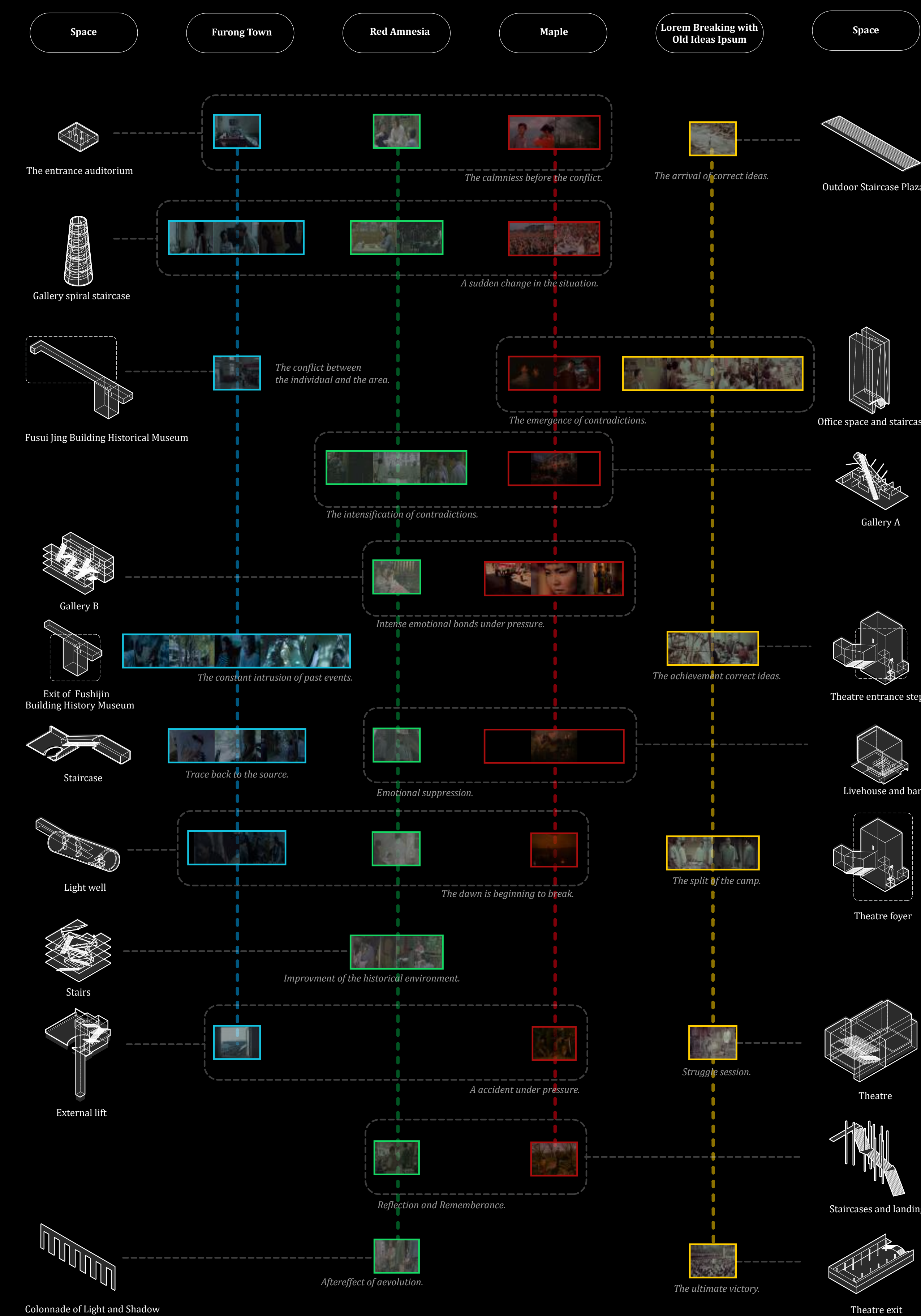
Be Saved

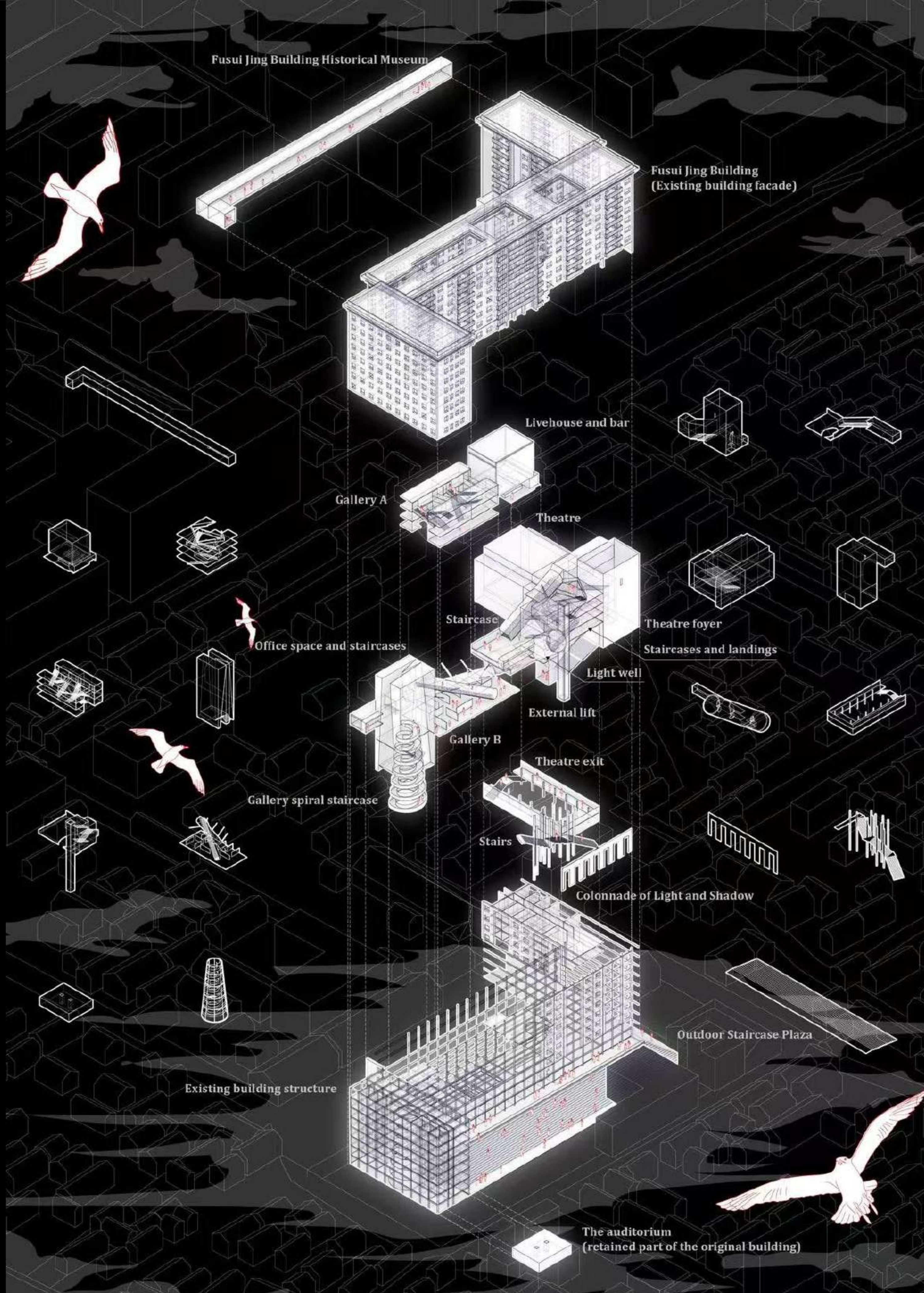
In 2019, the Fusui Jing Building was officially listed as a historical protected structure. To rescue this abandoned edifice and inject new vitality into it, I aspire to transform it into a new public building through adaptive renovation. While enabling it to serve new functions, the project also aims to revive and encourage people to confront that once-taboo period of history embedded within its walls.

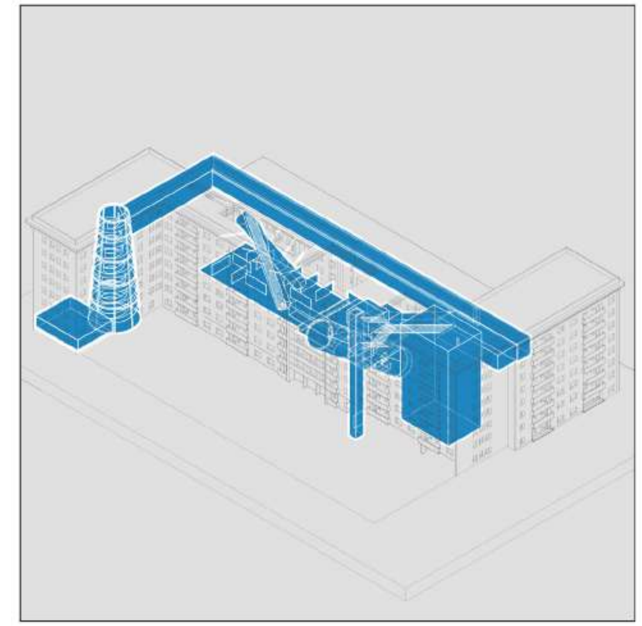
History

The unvarnished truth of history can never be fully discerned from a solitary vantage point; disparate subjective perspectives breed divergent versions of what people accept as fact. In a deliberate bid to draw closer to the authentic essence of history, I have incorporated four carefully selected films, with each work embodying a distinct, independent viewpoint. Through this multi-faceted narrative approach, I seek to piece together and reconstruct that long-overlooked, almost forgotten chapter of history.

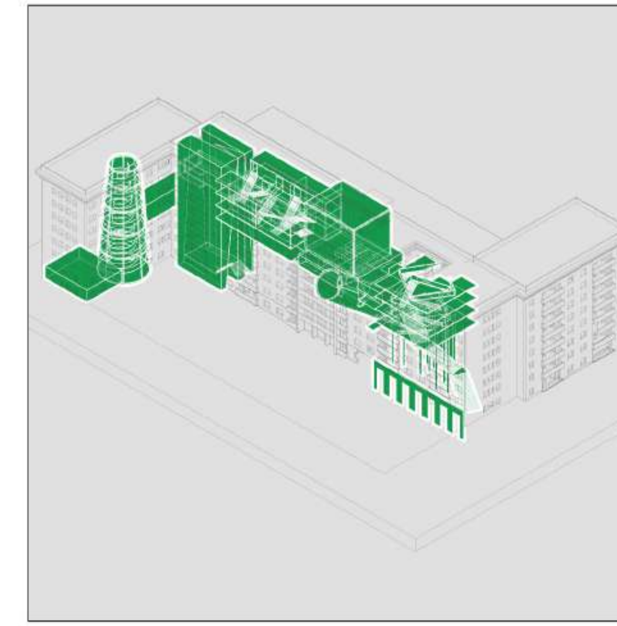




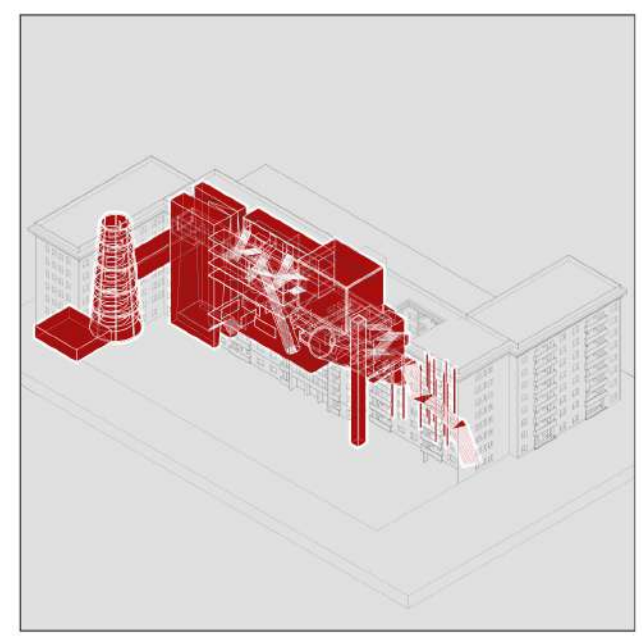




Route 1
History Museum



Route 2
Entertainment



Route 3
Exhibition

Route 4
Theater

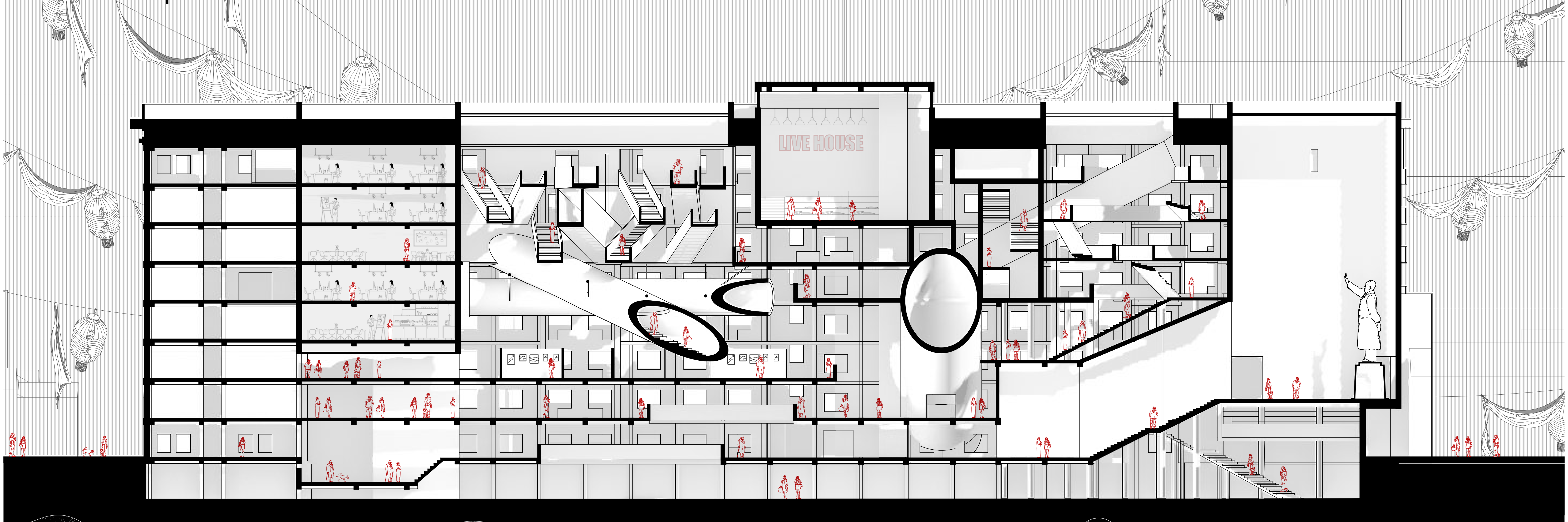
Interior Space

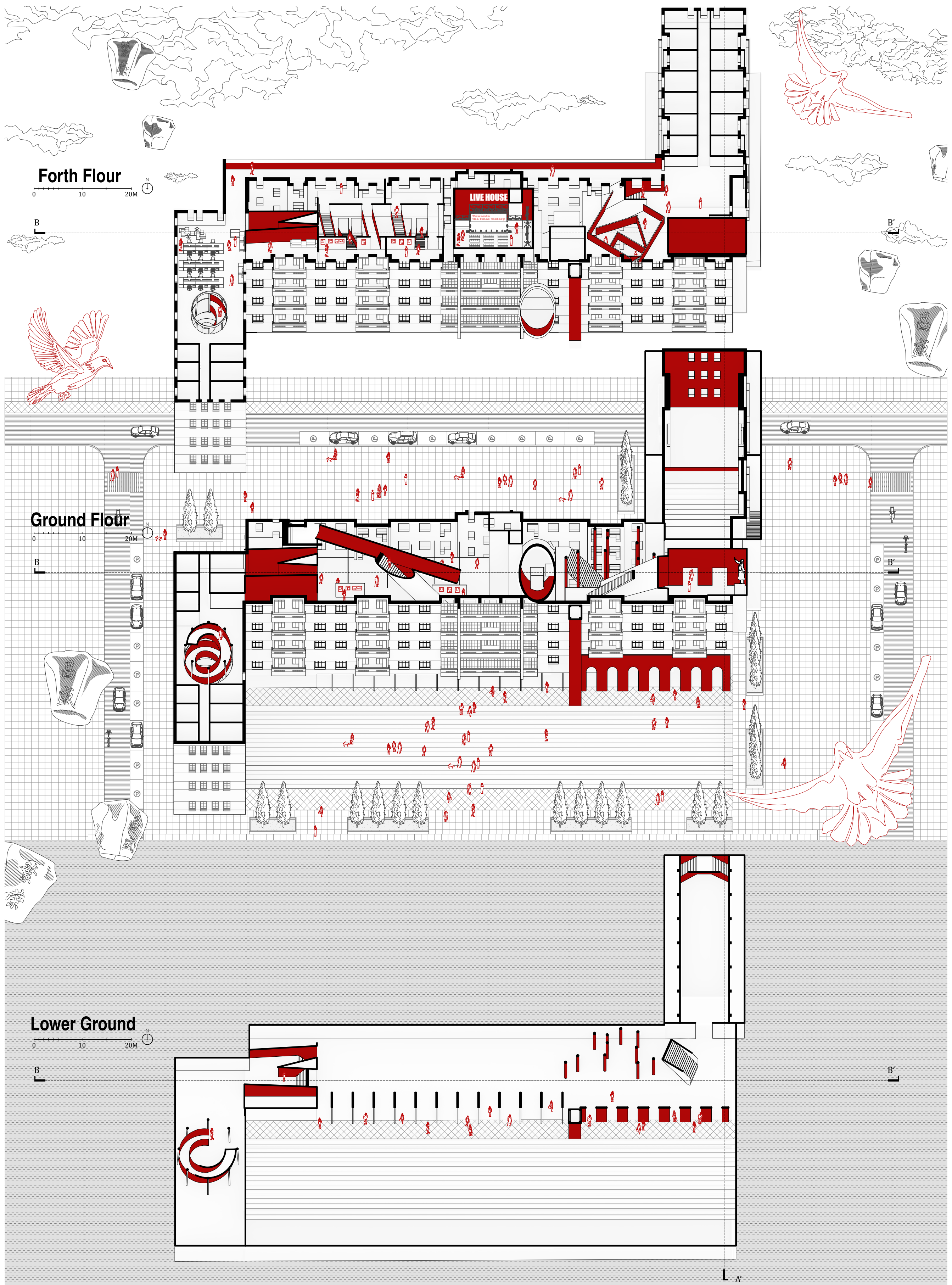
Spiral Stairs

Aerial Staircase

Exhibition

Theater Etrance

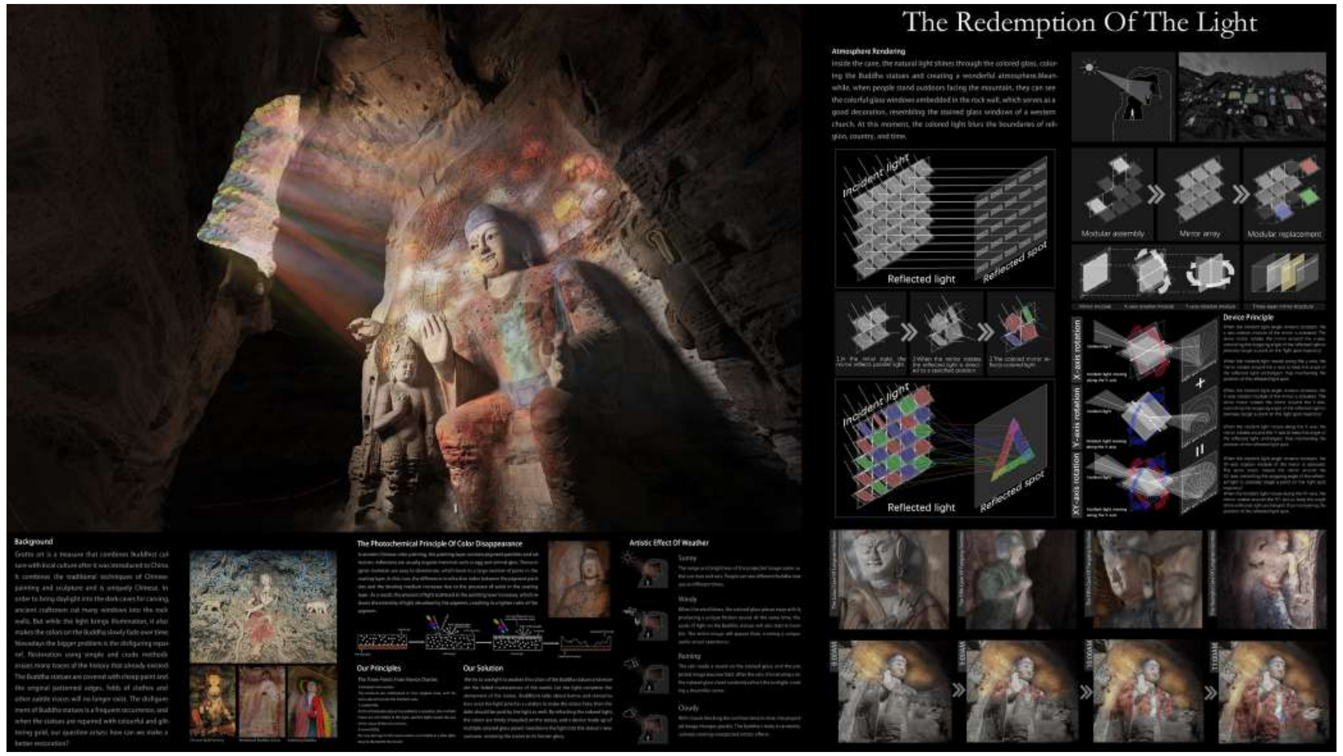




Theater Section

Front Facade

Other Works

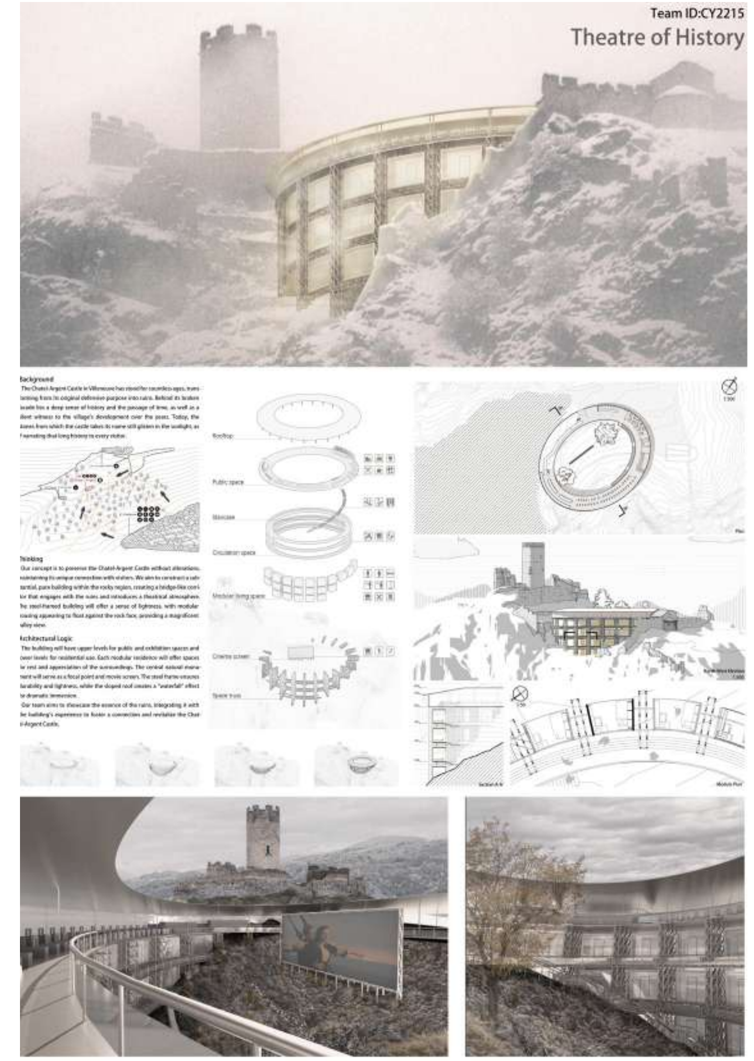


1."Velux" Competition

In this competition, I served as the team leader. I recruited team members, proposed the core design concept, allocated tasks, and supervised the progress. Additionally, I was responsible for drawing some analysis diagrams, drafting copy, and controlling the overall quality.

2.Handcrafted Model

This is an assignment for a hands-on course in building structures. As the leader of a four-person team, I guided my group members through structural design, digital model construction, and blueprint drafting. During the collaboration, I also took charge of the statistics and procurement of all model materials. To address the specific challenges in fabricating micro-models, I selected and purchased specialized tools for my team members.



3."Reuse the Fortress" Competition

In the "Reuse the Fortress" competition, as a team member, I collaborated with partners on conceiving architectural concepts, form design, housing layout, technical drawing, and copywriting.

4."Digital Future 2025" Workshop

During my participation in the World Model Group, I systematically learned the core knowledge of LLMs, VLMs, and VLAs. I conducted practical work on Qwen2.5 (including environment setup, weight downloading, and multimodal demo running) and mastered VLA architecture and diffusion models based on the GR00T N1 paper, while also grasping relevant technical concepts through code configuration.

In the training phase, I analyzed robot datasets, sorted out model architectures, finalized training workflows, studied key hyperparameters and their impact on loss, and completed formal training. Finally, I learned the VLA evaluation process, assessed results using ManiSkill.